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Iwase et al.

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(54) **DRIVER HOLDER, DRIVER MODULE, AND HEADSET**

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CPC **H04R 1/1016** (2013.01)

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H04R 1/1083; H04R 2460/01
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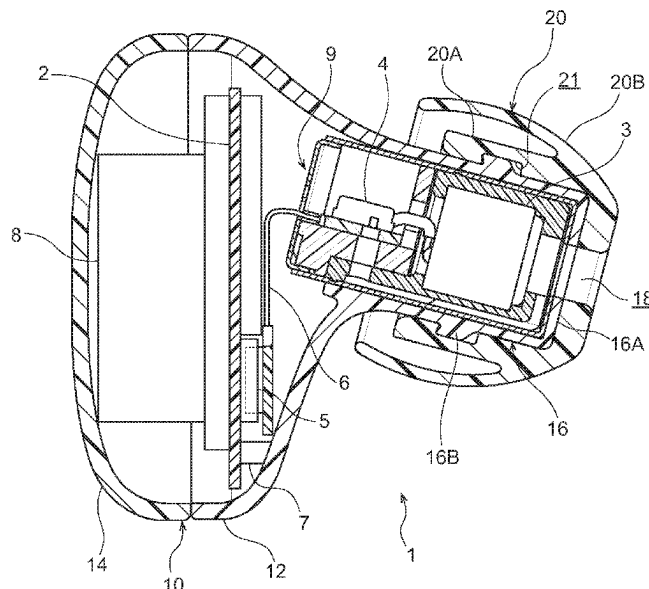
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(57) **ABSTRACT**

A driver holder is disposed inside a housing. This driver holder has a tubular portion formed in a tubular shape, and a driver is accommodated inside the tubular portion. Further, the tubular portion is press-fitted into a module case arranged inside the housing. Here, an opening at one end side in an axial direction of the tubular portion configures a first sound path that communicates the driver with the exterior of the housing. Further, a groove formed at a surface of the tubular portion configures a second sound path which is communicated with the first sound path and extends to a microphone arranged outside the tubular portion.

20 Claims, 19 Drawing Sheets



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FIG.1

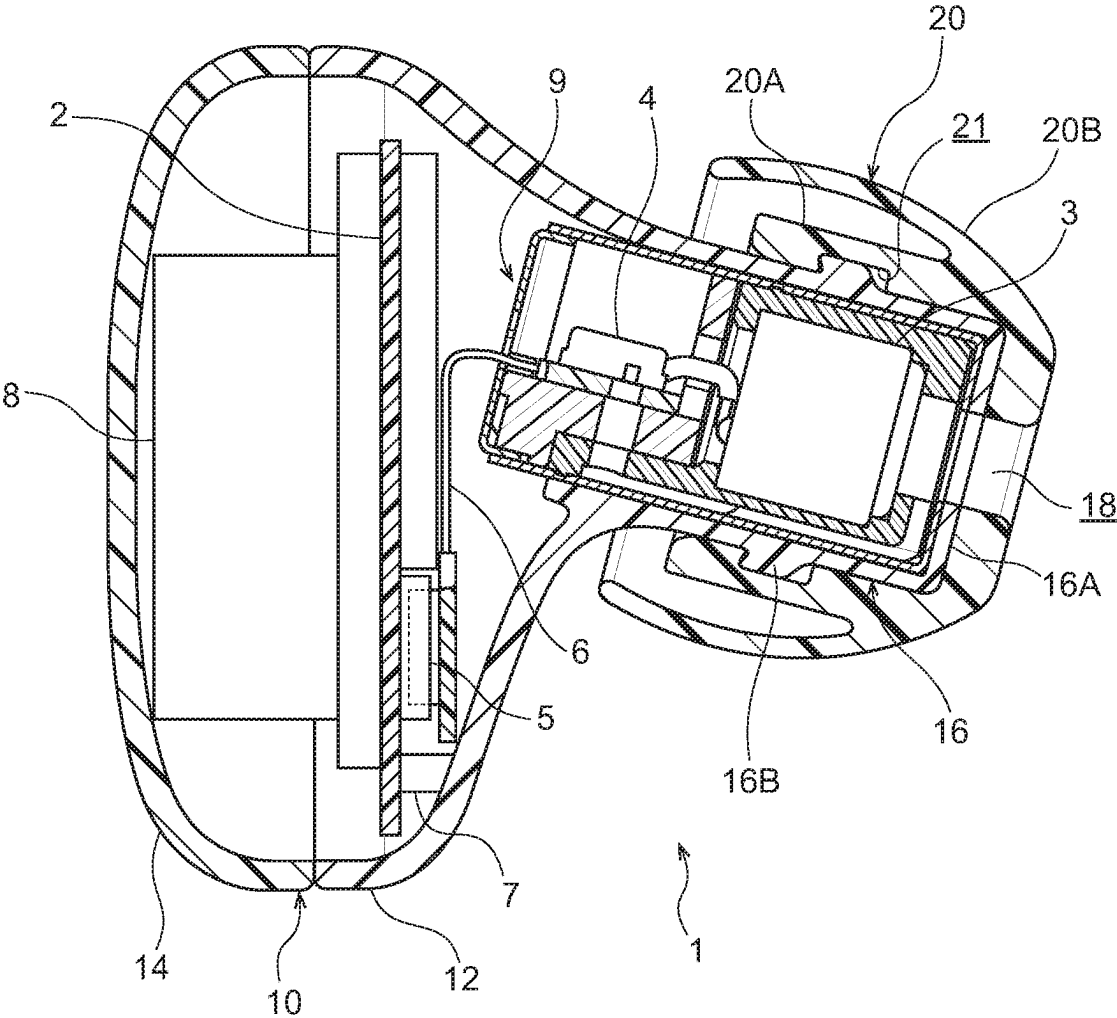


FIG.2

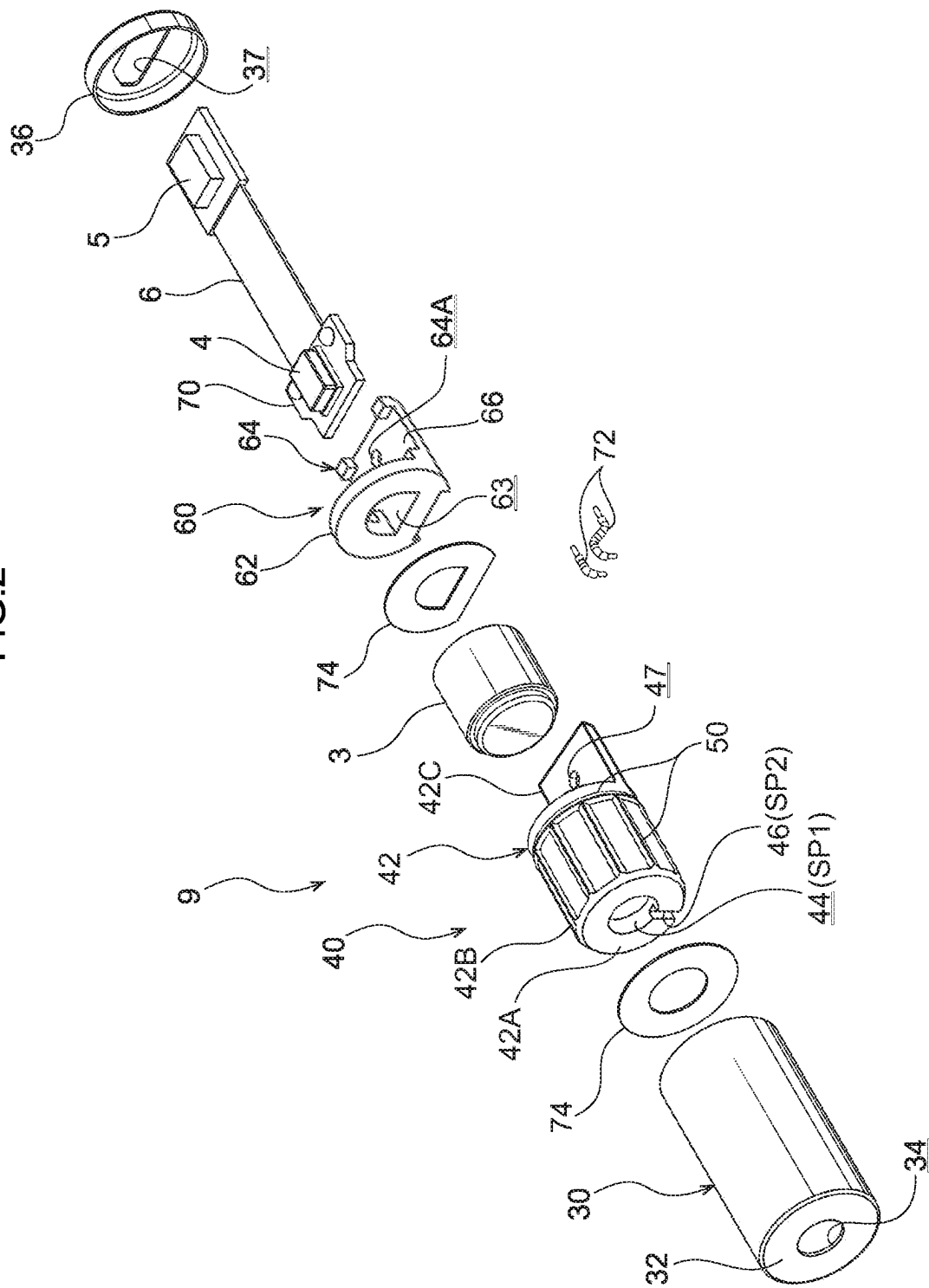


FIG.3

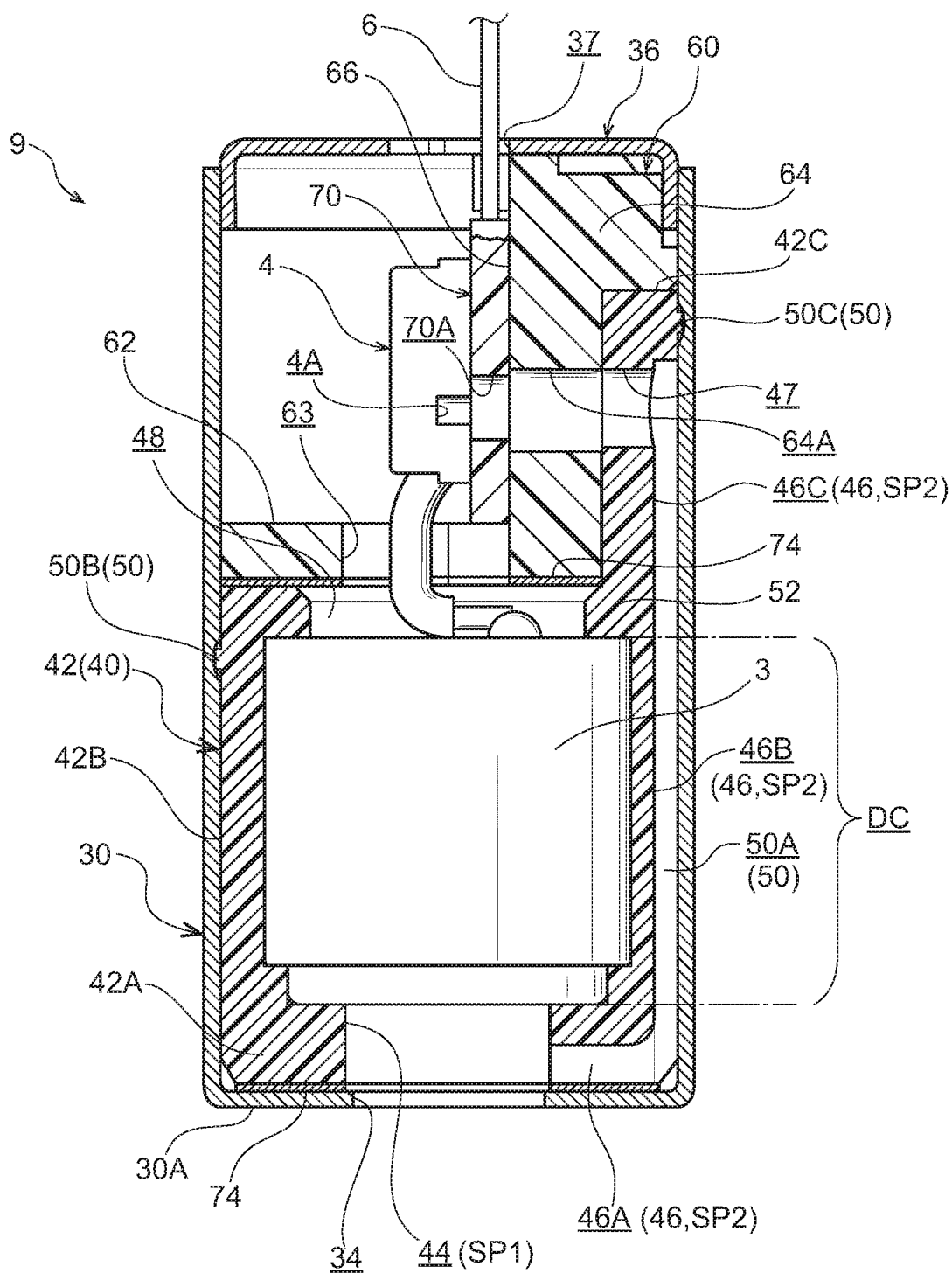


FIG.4A

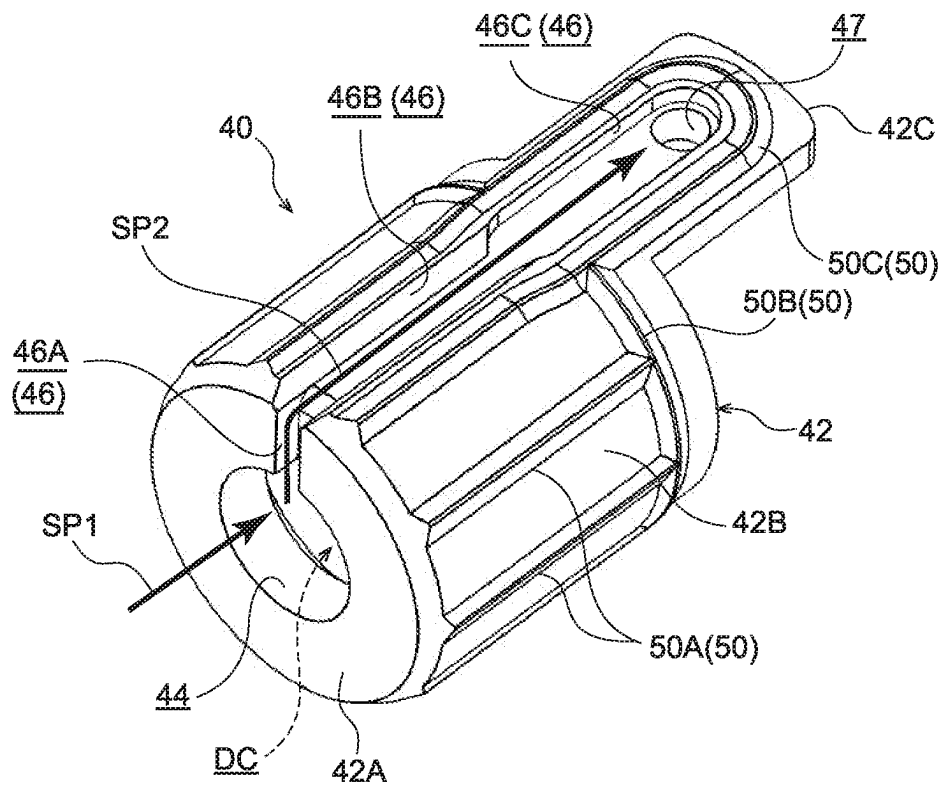


FIG.4B

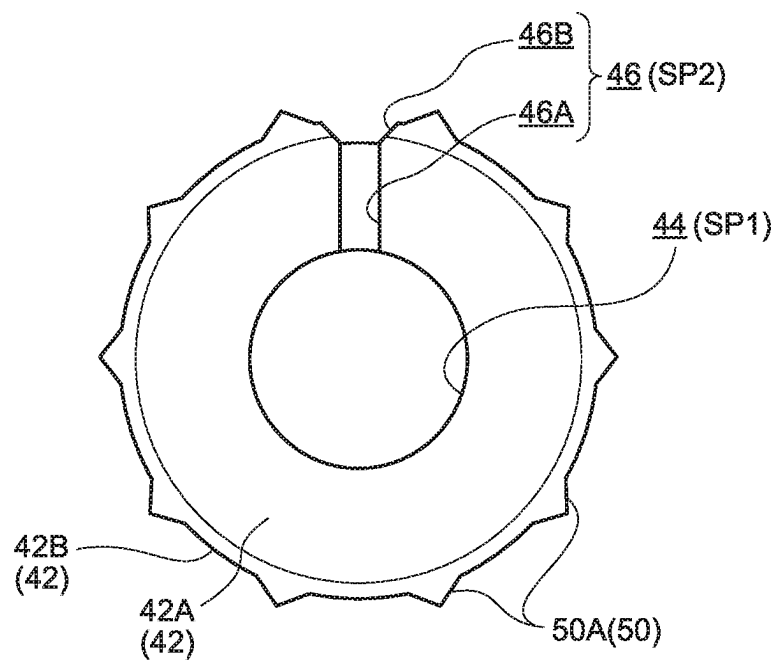


FIG.5A

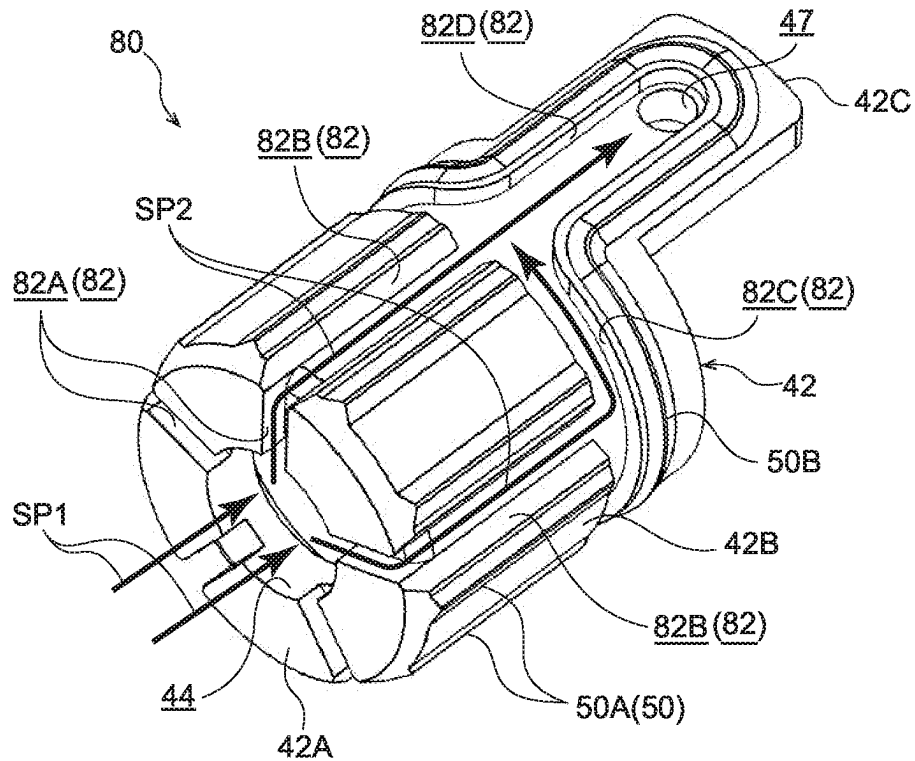


FIG.5B

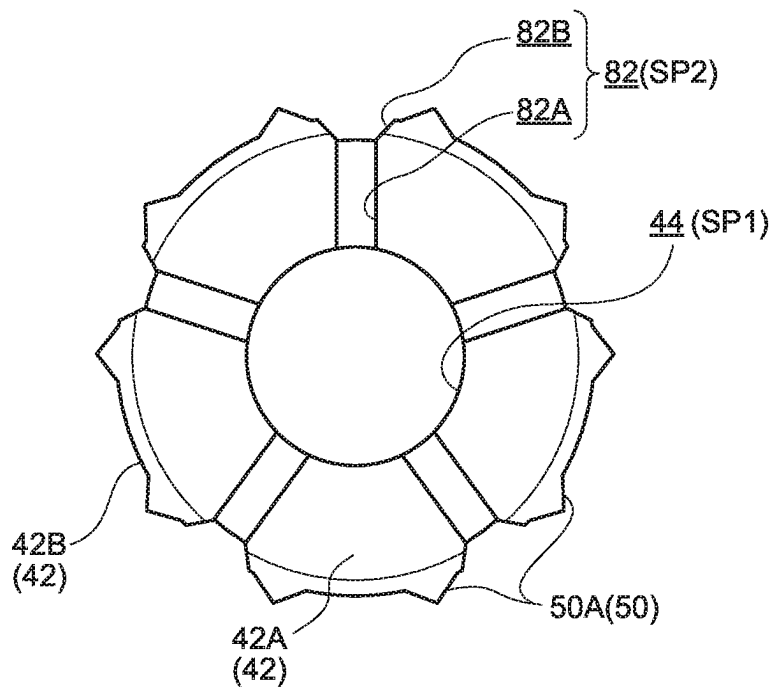


FIG.6

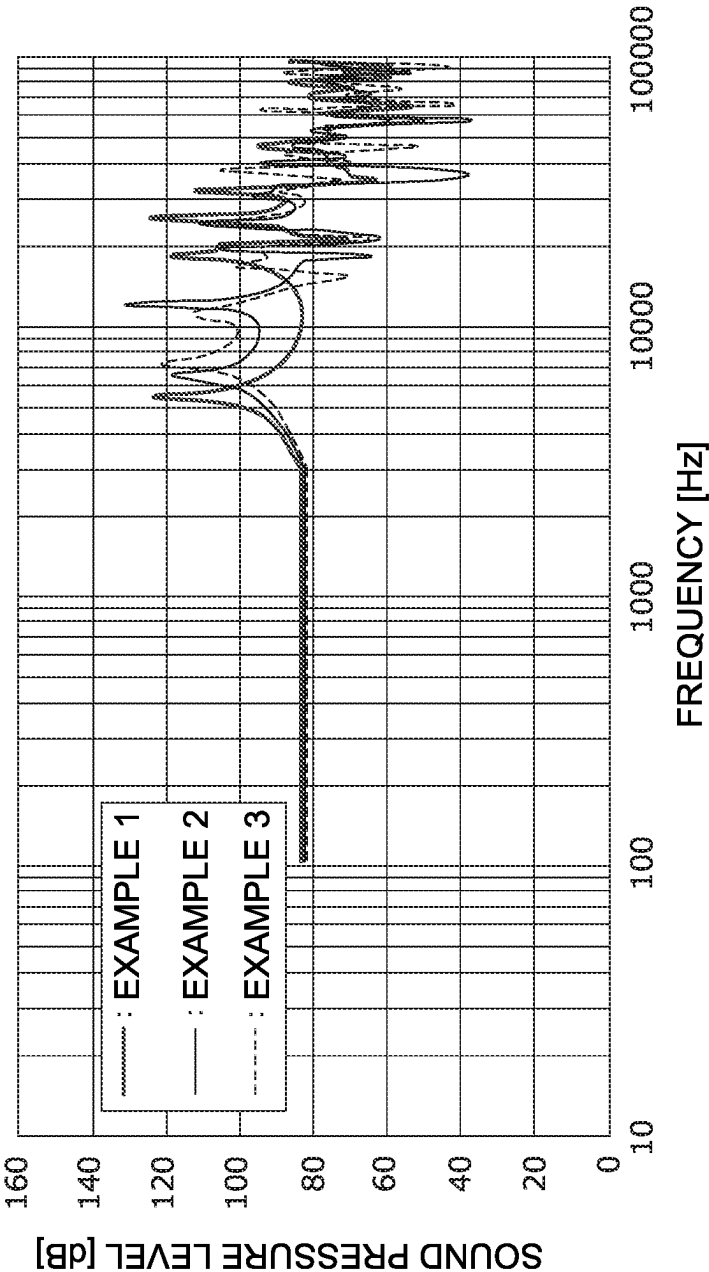


FIG. 7

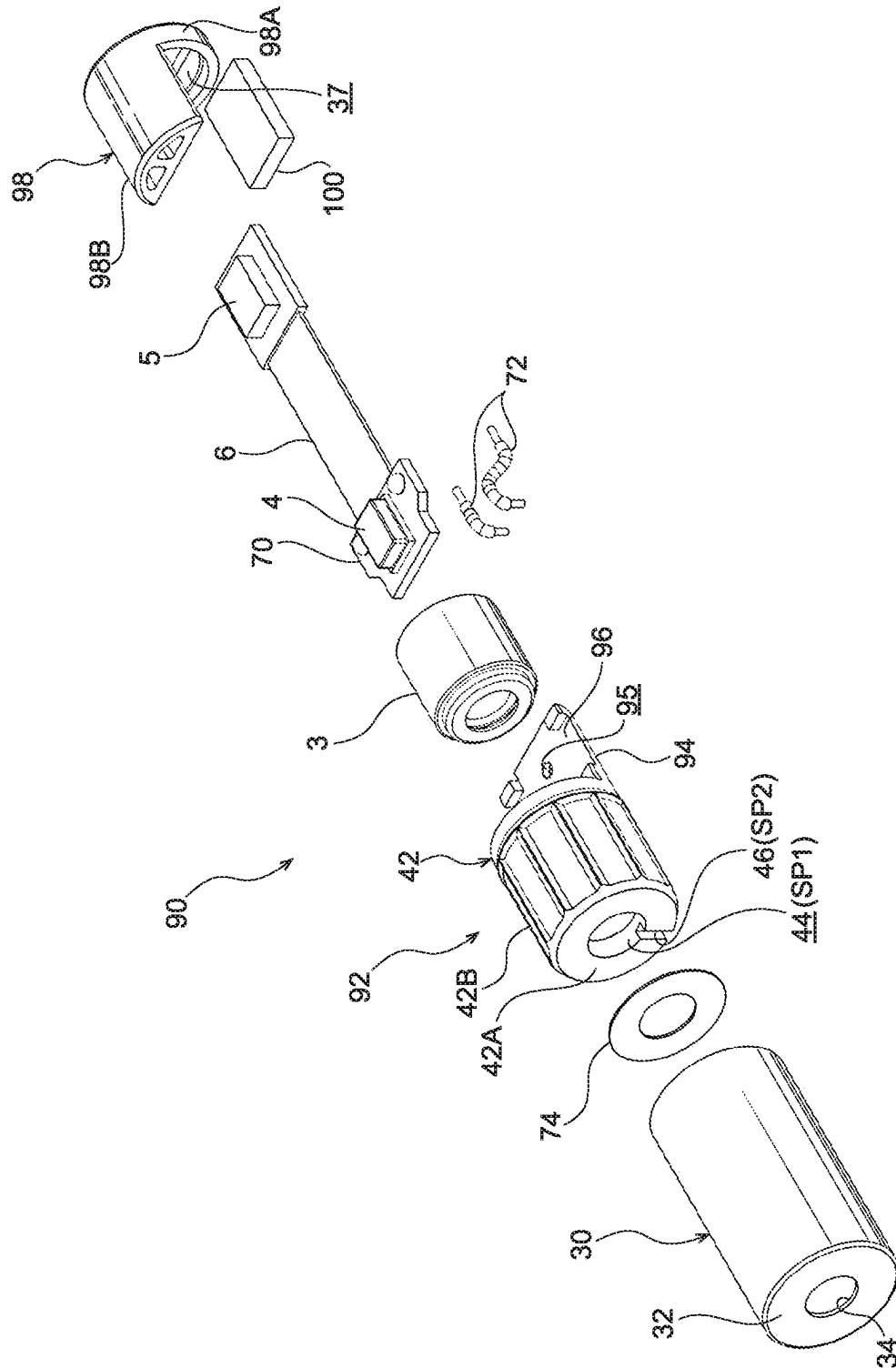


FIG.8

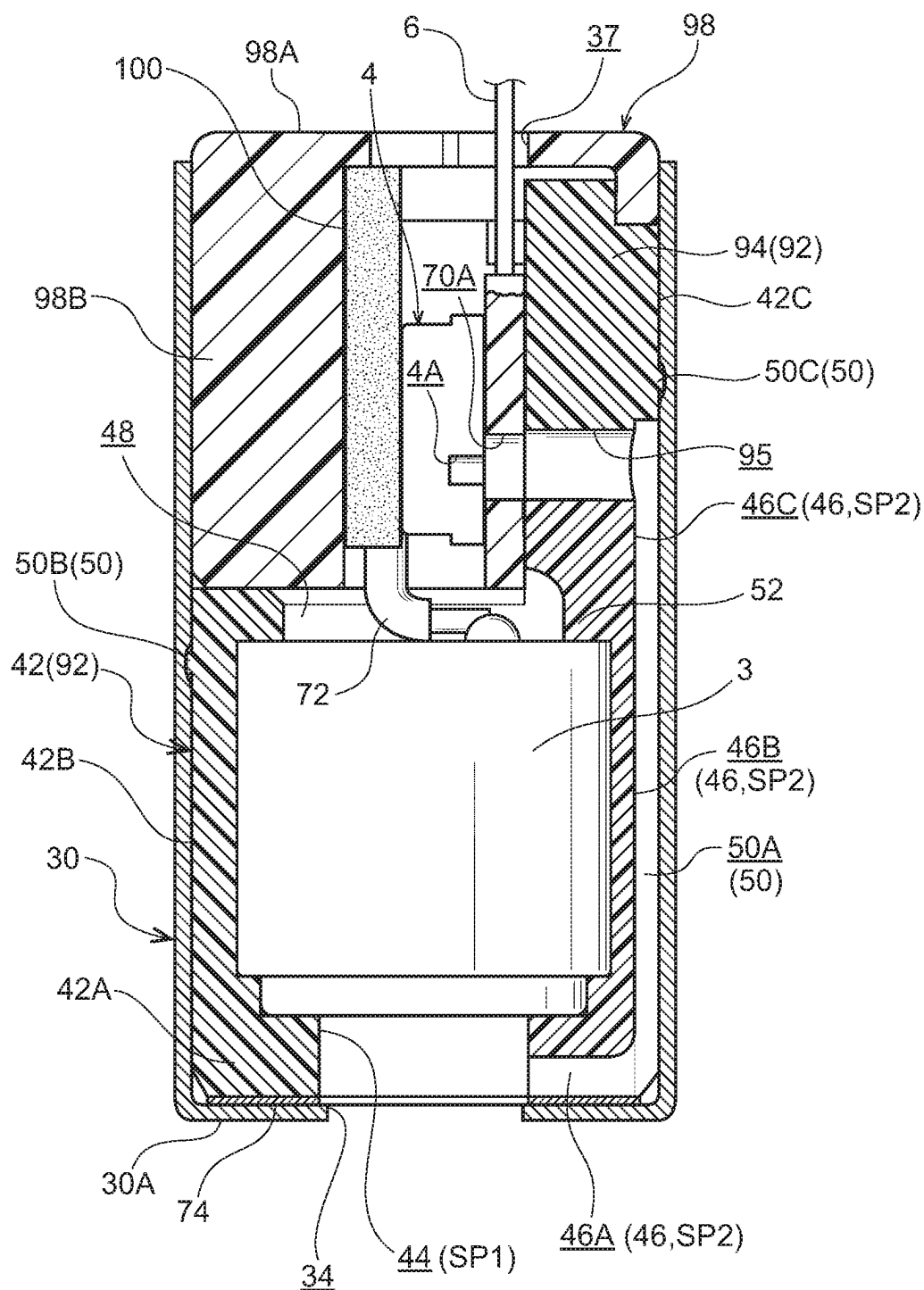
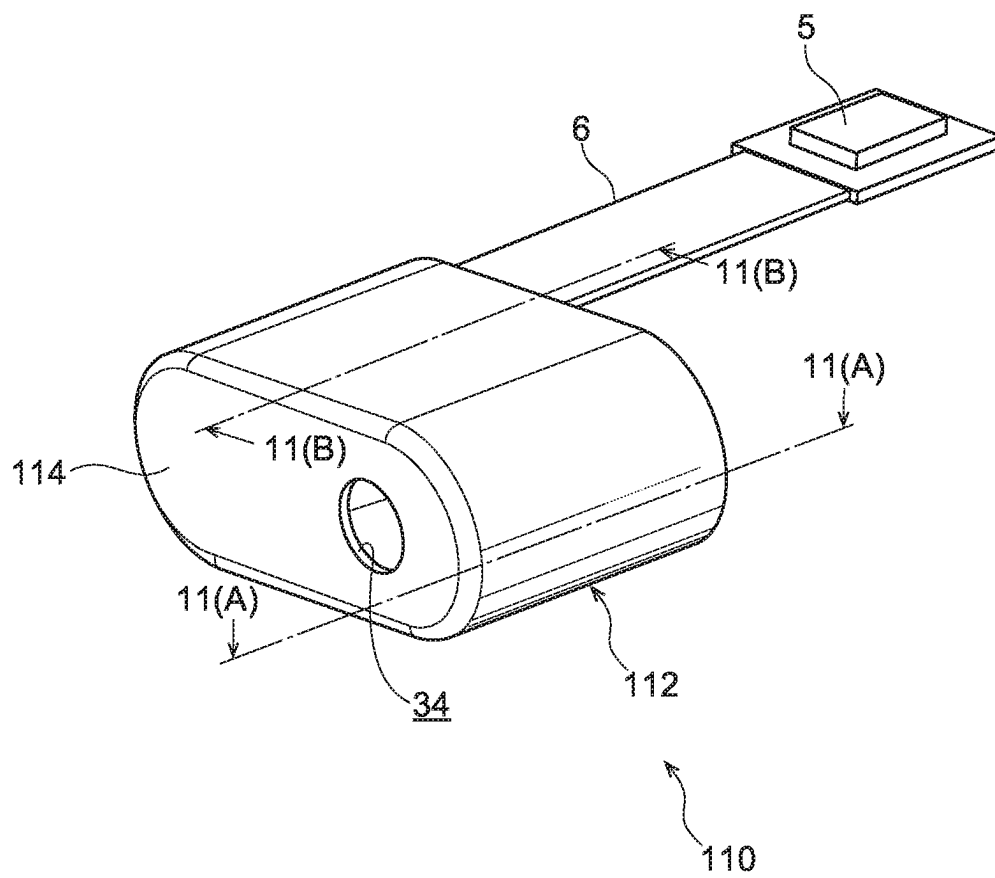


FIG.9



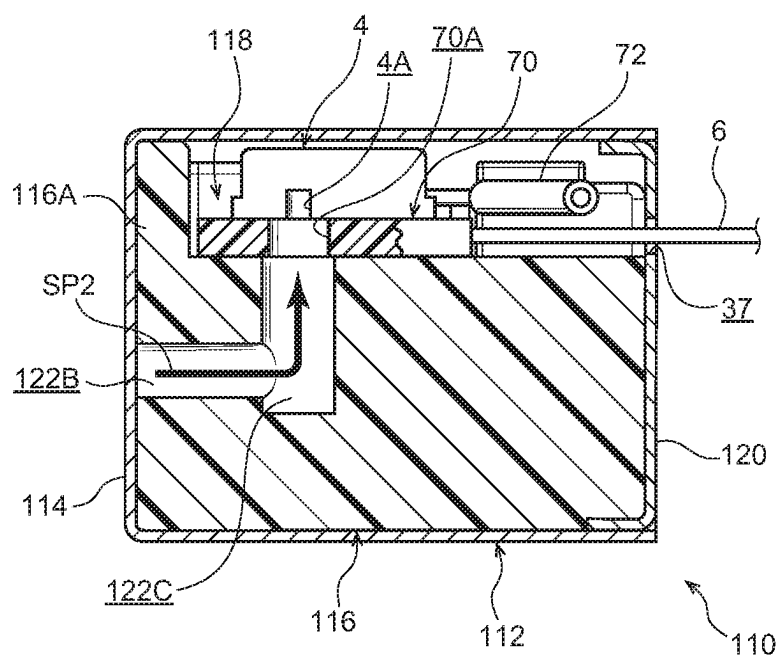


FIG.12A

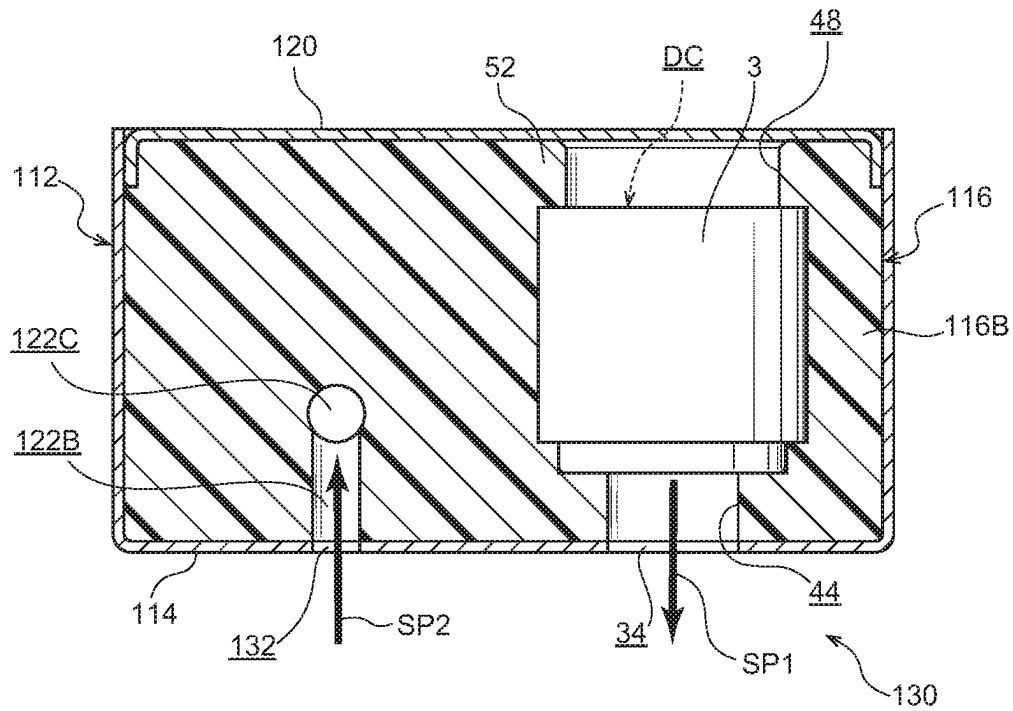


FIG.12B

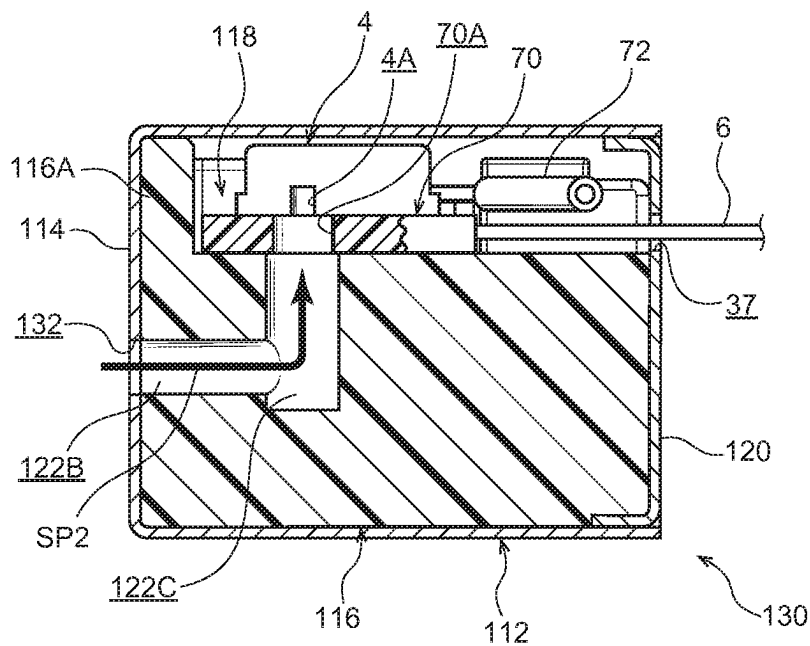


FIG. 13

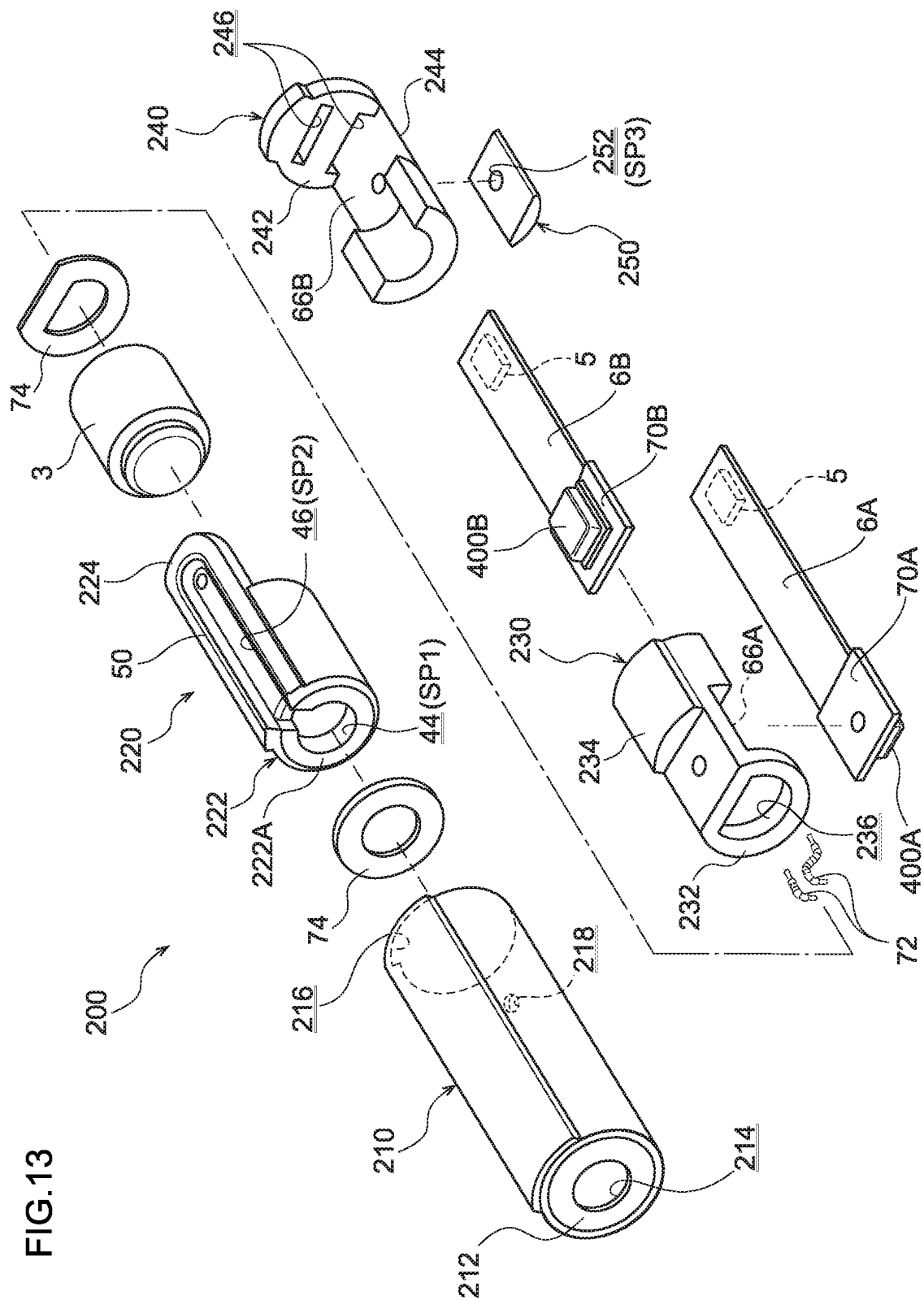


FIG. 14

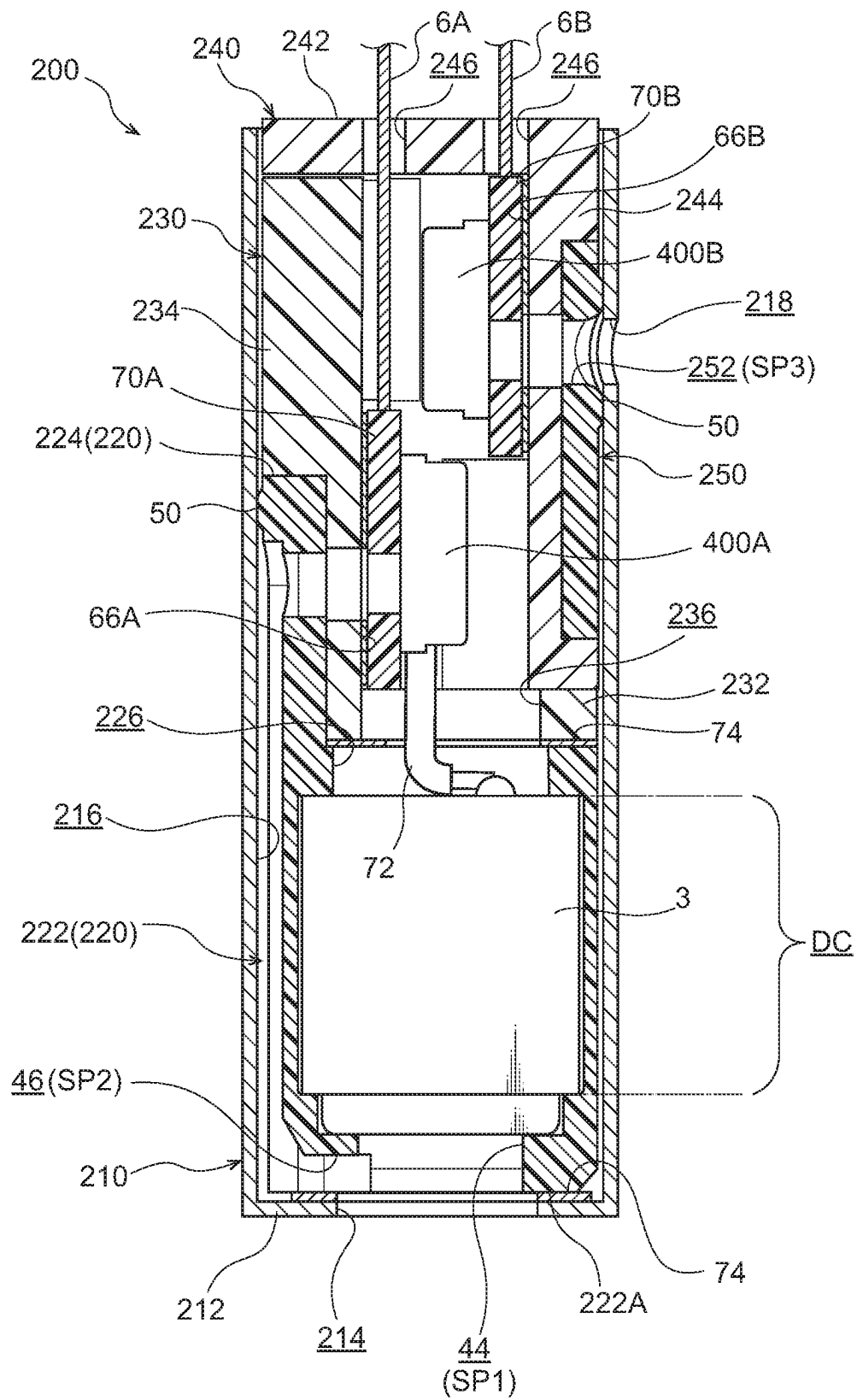


FIG.16

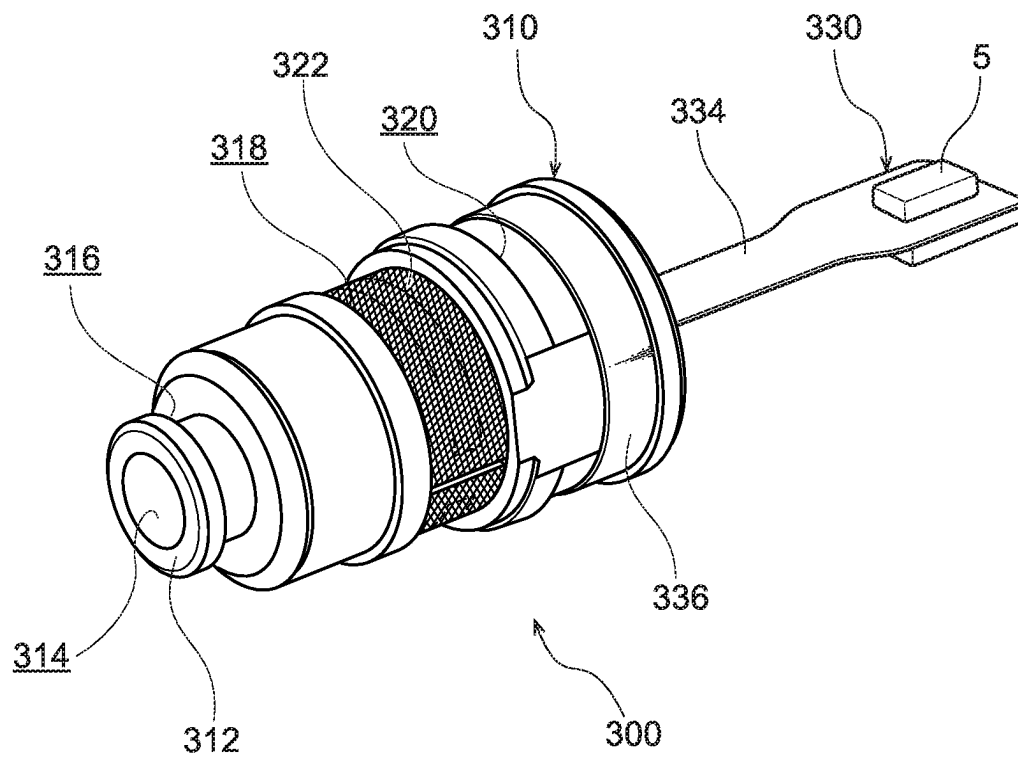


FIG.17

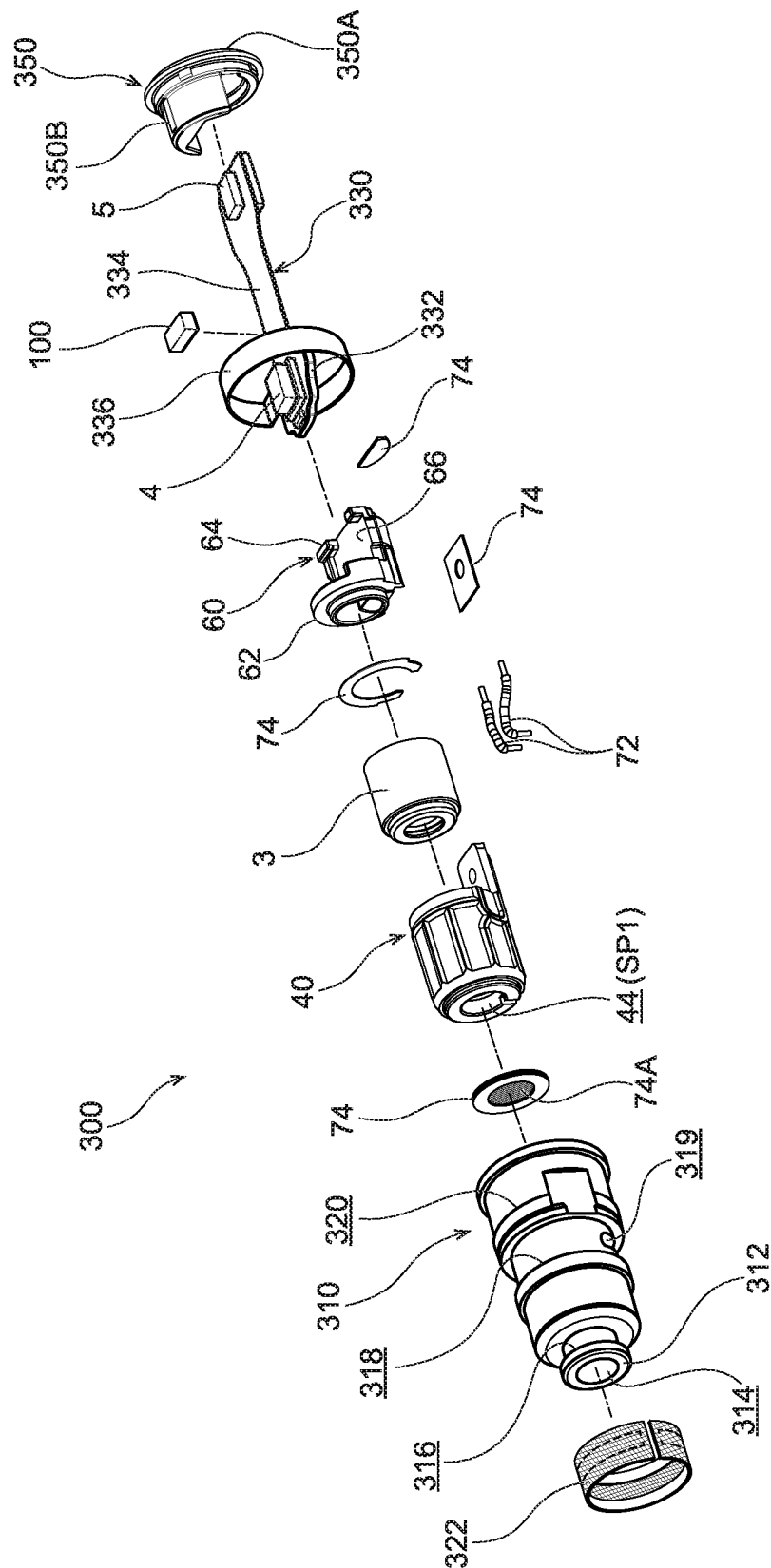


FIG.18

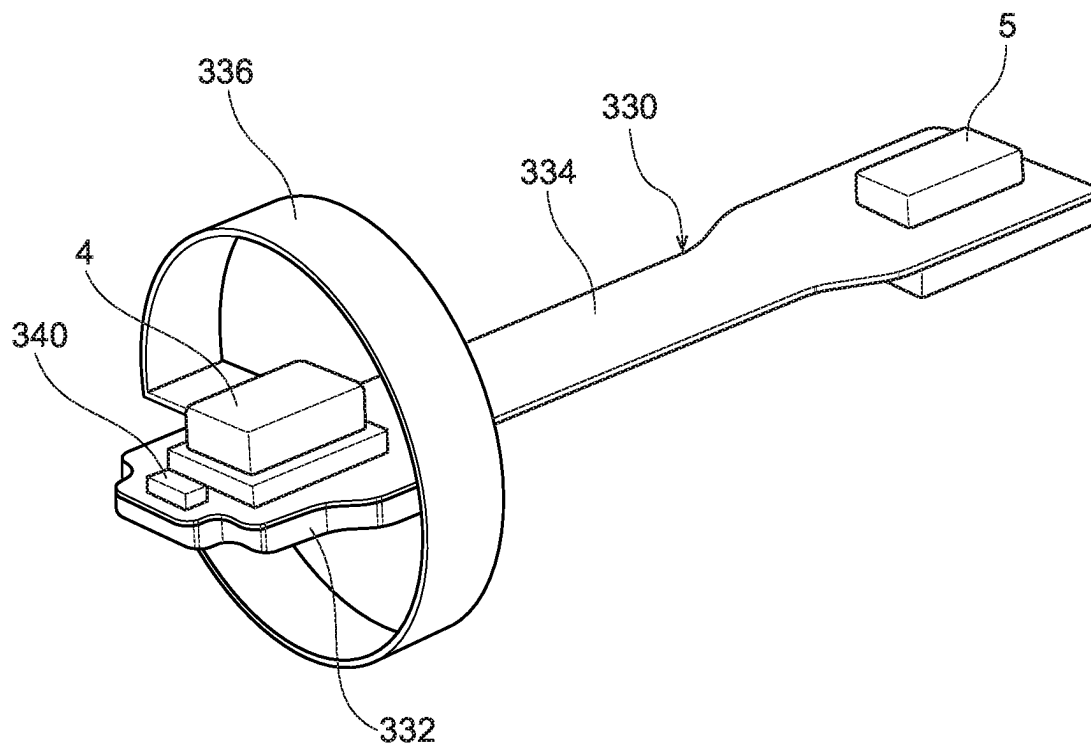
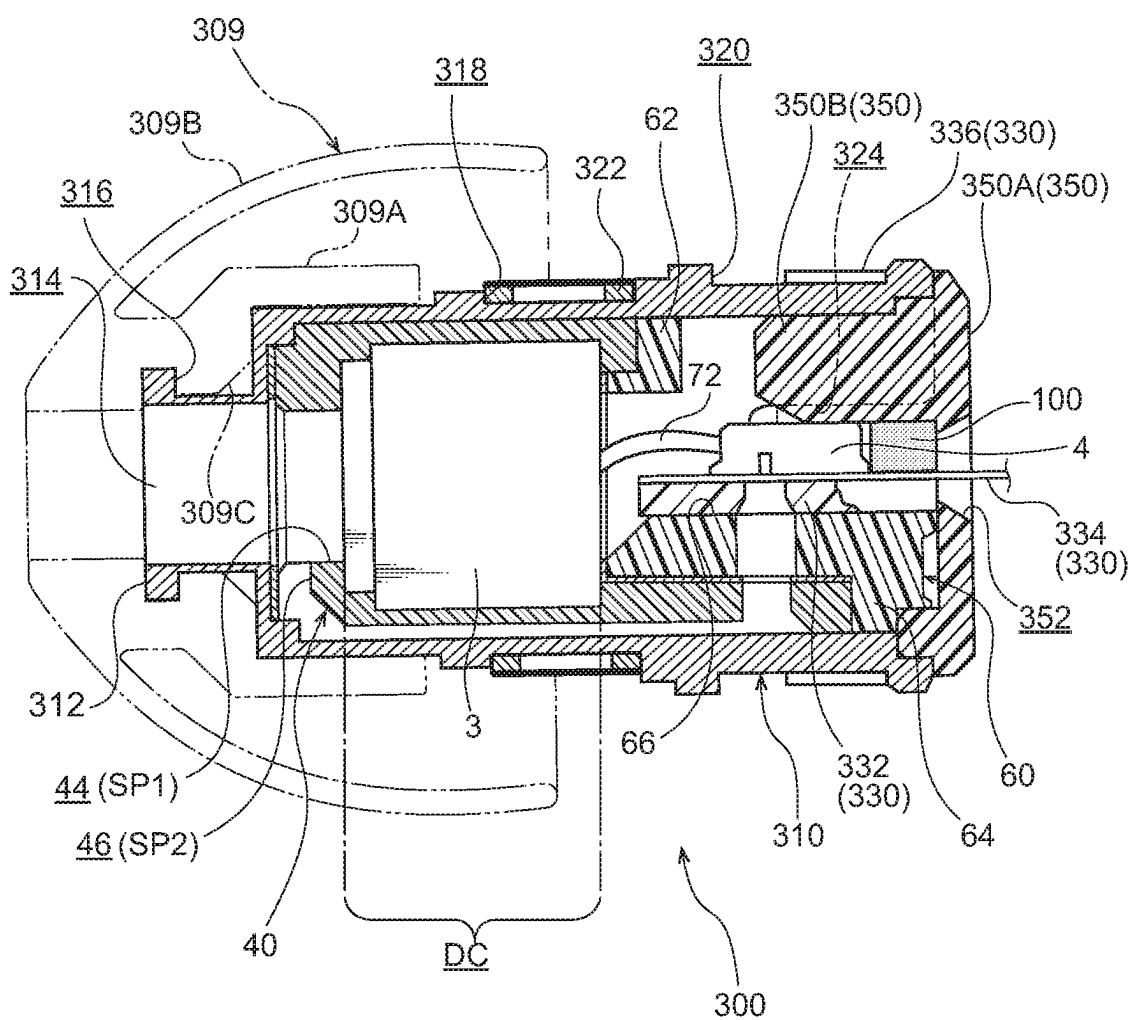


FIG. 19



**DRIVER HOLDER, DRIVER MODULE, AND
HEADSET**

TECHNICAL FIELD

The present disclosure relates to a driver holder, a driver module and a headset.

BACKGROUND ART

Japanese Patent Application Laid-Open (JP-A) No. 2015-126509 discloses a technique of providing a partition portion inside an ear tip that is a part of a headset housing and that corresponds to an ear canal insertion portion, and forming a sound path for a microphone at an outer side of a sound path for a driver. These sound paths respectively communicate the inside of the user's ear canal with a driver accommodation portion and a microphone accommodation portion provided inside the housing.

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

However, with the technique described in JP-A No. 2015-126509, since the sound path for the microphone is formed by the housing, when attempting to modify the sound path in order to optimize the acoustic characteristics, there is the problem that it cannot be easily modified.

Further, although the sound path for the microphone needs to be sealed, with the technique described in JP-A No. 2015-126509, since multiple parts are assembled to form a housing, there is the problem that it is not easy to effect assembly while ensuring the sealing properties of the sound path. Further, the positional relationship between the driver and the microphone is not stable, and there is the problem that variations due to assembly are likely to occur.

The present disclosure, in consideration of the foregoing problems, aims to provide a driver holder, a driver module, and a headset that facilitate modification of the sound path for a microphone and acquisition of sealing properties, and that exhibit improved ease of assembly.

Means for Solving the Problem

A driver holder according to a first aspect is a driver holder for holding a driver at an interior of a hollow housing, the driver holder including: a tubular portion formed in a tubular shape, the driver being accommodated at an inner side thereof, and the tubular portion being configured to be press-fitted into a holder insertion portion disposed inside the housing; a first sound path configuring an opening at one end side in an axial direction of the tubular portion, and communicating the driver with an exterior of the housing; and a second sound path configured by a groove formed at a surface of the tubular portion, the second sound path being communicated with the first sound path and extending as far as a microphone disposed at an outer side of the tubular portion.

According to the first aspect, the driver holder is disposed inside the housing. This driver holder has a tubular portion formed in a tubular shape, and a driver is accommodated inside the tubular portion. Further, the tubular portion is configured to be press-fitted into a holder insertion portion inside the housing. As a result, the driver arranged inside the housing is held by the driver holder.

Here, an opening at one end side in an axial direction of the tubular portion configures a first sound path that communicates the driver with the housing exterior. As a result, an acoustic signal output by the driver is radiated to the exterior via the first sound path.

A second sound path is formed at a surface of the tubular portion. This second sound path is configured by a groove formed at a surface of the tubular portion, is communicated with the first sound path, and extends to a microphone arranged at an outer side of the tubular portion. As a result, the acoustic signal propagates to the microphone via the first sound path and the second sound path.

According to the foregoing configuration, it is sufficient to modify the respective sound paths by modifying the shape of the driver holder, which is a simple matter in that no design change of the housing is required. In addition, with these sound paths, each sound path is put into a sealed state by press-fitting the tubular portion into the holder insertion portion. Therefore, assembly while ensuring the hermeticity of the second sound path for the microphone is facilitated. Furthermore, since the second sound path is provided integrally with the driver holder, the positional relationship between the driver and the microphone is stabilized. Therefore, variations due to assembly can be suppressed.

A driver holder according to a second aspect is the configuration according to the first aspect, in which the driver holder is formed with an elastic material.

In the second aspect, since the driver holder is formed with an elastic material, the tubular portion can be easily press-fitted into the holder insertion portion by elastic deformation thereof, and the ease of assembly of the driver holder can be improved.

A driver holder according to a third aspect is the configuration according to the first aspect or the second aspect, in which a rib projecting from the surface is formed at the surface of the tubular portion, and the tubular portion is press-fitted into the holder insertion portion while elastically deforming the rib.

According to the third aspect, the tubular portion is press-fitted into the holder insertion portion while elastically deforming the rib. This reduces the frictional resistance when the tubular portion is press-fitted into the holder insertion portion, thereby improving the workability during manufacturing.

A driver holder according to a fourth aspect is the configuration according to any one of the first to third aspects, in which a driver pressing portion is provided at the tubular portion, the driver pressing portion protruding toward an inner side in a radial direction from an opening at another end side in the axial direction at which the driver is inserted, and the driver is accommodated inside the tubular portion by surmounting the driver pressing portion, and is pressed toward one side in the axial direction by elastic force of the driver pressing portion.

According to the fourth aspect, the driver is pressed toward one side in the axial direction of the tubular portion by the elastic force of the driver pressing portion. Therefore, the sealing performance between the driver and the tubular portion is effectively enhanced in the vicinity of the first sound path, which in turn contributes to improvement of the sealing performance of the first sound path. Further, in the manufacturing process, by surmounting the driver pressing portion, since the operator can clearly confirm that the driver is housed in an appropriate position in the tubular portion by visual and tactile sense, product assembly accuracy and workability are improved.

A driver holder according to a fifth aspect is the configuration according to any one of the first to fourth aspects, in which plural of the grooves are formed at the surface of the tubular portion.

According to the fifth aspect, plural grooves forming the second sound path are formed at the surface of the tubular portion. As a result, the impact absorption of the driver holder can be enhanced, as a result of which the impact resistance of the driver is improved. In addition, it is possible to form an optimum sound path for the microphone matching the acoustic characteristics while also reducing the outer diameter of the driver holder.

A driver holder according to a sixth aspect is the configuration according to any one of the first to fifth aspects, in which the holder insertion portion is configured by a bottomed tubular module case, and the tubular portion is press-fitted inside the module case.

According to the sixth aspect, since the holder insertion portion is configured by a bottomed tubular module case, by press-fitting the driver holder into the module case, the driver and sound path can be easily modularized.

A driver holder according to a seventh aspect is the configuration according to the sixth aspect, in which the module case has another end side in the axial direction closed off by a closing member, and a holder pressing portion is provided at another end side in the axial direction of the tubular portion, the holder pressing portion being disposed between the closing member and the tubular portion, and pressing the tubular portion toward the one end side in the axial direction.

According to the seventh aspect, a holder pressing portion is arranged between the closing member of the module case and the tubular portion, and presses the driver holder toward the one end side in the axial direction. Therefore, by putting the module case into a closed state, the tubular portion can be pressed toward the one end side in the axial direction, and therefore, it is possible to stabilize the positional relationship between the driver and the microphone and to reduce variations due to assembly.

A driver holder according to an eighth aspect is the configuration according to the seventh aspect, in which the holder pressing portion is provided with a microphone fixing portion, which fixes a microphone, at a face at an opposite side from a face that contacts the tubular portion.

In the driver holder according to the eighth aspect, the holder pressing portion is provided with a microphone fixing portion, and a microphone is disposed inside the module case. As a result, the main functional parts that configure the headset, such as the driver, microphone, and sound paths, are modularized, and product management is superior because the product can be managed as an assembly. Further, by assembling the main components in advance, the installation work on the housing side can be simplified. Furthermore, since the module case can be used in combination with housings for plural products having different shapes, versatility is excellent.

Since a driver module according to a ninth aspect includes the driver holder according to the first to eighth aspects, as described above, it is possible to easily change the sound path for the microphone and ensure sealing, and to improve the assembly efficiency.

Since a headset according to a tenth aspect includes the driver holder according to the first to eighth aspects, as described above, it is possible to easily change the sound path for the microphone and ensure sealing, and to improve the assembly efficiency.

According to the present disclosure, a driver holder, a driver module, and a headset that facilitate modification of the sound path for a microphone and acquisition of sealing properties, and that exhibit improved ease of assembly, are provided.

BRIEF EXPLANATION OF THE DRAWINGS

FIG. 1 is a cross-sectional view showing the overall configuration of a headset according to a first embodiment.

FIG. 2 is an exploded perspective view of a driver module according to the first embodiment.

FIG. 3 is an axial direction longitudinal sectional view of the driver module according to the first embodiment.

FIG. 4A is a perspective view of a driver holder according to the first embodiment.

FIG. 4B is a diagram of the driver holder as viewed from the front side in the axial direction.

FIG. 5A is a perspective view of a driver holder according to a second embodiment.

FIG. 5B is a diagram of the driver holder as viewed from the front side in the axial direction.

FIG. 6 is a graph relatively showing frequency characteristics of the driver at the microphone position in accordance with variations of the second sound path.

FIG. 7 is an exploded perspective view of a driver module according to a third embodiment.

FIG. 8 is an axial direction longitudinal sectional view of the driver module according to the third embodiment.

FIG. 9 is a perspective view of a driver module according to a fourth embodiment.

FIG. 10 is an exploded view of the driver module according to the fourth embodiment.

FIG. 11A is a cross-sectional plan view along line 11A-11A in FIG. 9.

FIG. 11B is a longitudinal sectional view taken along line 11B-11B in FIG. 9.

FIG. 12A is a cross-sectional plan view of a driver module according to a fifth embodiment, corresponding to FIG. 11A.

FIG. 12B is a longitudinal sectional view of the driver module according to the fifth embodiment, corresponding to FIG. 11B.

FIG. 13 is an exploded perspective view of a driver module according to a sixth embodiment.

FIG. 14 is an axial direction longitudinal sectional view of the driver module according to the sixth embodiment.

FIG. 15 is a cross-sectional view showing the overall configuration of a headset according to a seventh embodiment.

FIG. 16 is a perspective view of a driver module according to a seventh embodiment.

FIG. 17 is an exploded perspective view of the driver module according to the seventh embodiment.

FIG. 18 is a perspective view of a substrate assembly according to the seventh embodiment.

FIG. 19 is an axial direction longitudinal sectional view of the driver module according to the seventh embodiment.

BEST MODE FOR CARRYING OUT THE INVENTION

First Exemplary Embodiment

A headset 1 according to a first embodiment is described below with reference to FIGS. 1 to 4. In the present

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specification, for convenience of explanation, the eardrum side of the external auditory canal (ear canal) when the headset **1** is worn is referred to as the front side, and the entrance side or auricle side of the ear canal is referred to as the rear side.

Overall Configuration

As shown in FIG. 1, the headset **1** has a hollow housing **10** accommodating functional components inside. The housing **10** is formed by fitting together a front housing **12** and a rear housing **14**.

The front housing **12** is formed in a tubular shape having an outer shape of an oblique truncated cone as a whole, and gradually decreases in diameter from the rear side toward the front side. As a result, the front housing **12** has a shape protruding toward the eardrum side of the user.

An ear canal insertion portion **16** is provided at a front portion of the front housing **12** and protrudes from the top of the oblique truncated cone toward the eardrum side. The ear canal insertion portion **16** is formed in a bottomed tubular shape whose axial direction is the front-rear direction, and a housing opening **18** that communicates the inside and outside of the front housing **12** is provided in the center of a bottom surface **16A** that configures a front end part. Inside the ear canal insertion portion **16**, a driver module **9**, which is described below, is accommodated.

An earpiece **20** is attached to the outer periphery of the ear canal insertion portion **16**. The earpiece **20** is also called an eartip, an earpad, or an earcap, and consists of an elastic material such as silicone rubber. The earpiece **20** has a cylindrical portion **20A** fitted around the outer circumference of the ear canal insertion portion **16**. The cylindrical portion **20A** is configured such that a fitting groove **21** provided on the inner periphery fits onto a fitting projection **16B** provided on the outer periphery of the ear canal insertion portion **16**. At the leading end of the cylindrical portion **20A**, a hemispherical cover-like adhesion portion **20B** is integrally provided so as to be deployed covering the entire cylindrical portion **20A**. The adhesion portion **20B** is configured to closely contact the wall surface of the user's ear canal and to block the ear canal when the headset **1** is in use.

The rear housing **14** is formed in a shallow dish shape that opens forward and is arranged so as to close off the rear opening of the front housing **12**.

A printed circuit board **2** is arranged in a housing space defined by the rear portion of the front housing **12** and the rear housing **14**. The printed board **2** is a board on which electronic components necessary for controlling the headset **1** are mounted. For example, various controls are performed according to the purpose of use of the headset **1**, such as output control of, for example, output signals from a driver **3**, which is described below, adjustment of, for example, the sensitivity of a microphone **4**, and cancellation frequency bands and levels in noise canceling when performing personal authentication with the headset. The printed circuit board **2** is arranged in an orientation in which the board thickness direction is substantially in the front-rear direction, and has mounting surfaces on both the front and back surfaces. In the present embodiment, a flexible board **6** connected to a driver module **9** via a connector **5** and a charging terminal **7** of the headset **1** are connected to the front mounting surface of the printed board **2**. Further, a battery **8** is arranged behind the printed circuit board **2**.

Driver Module

Next, the configuration of the driver module **9** is described in detail with reference to FIGS. 2 to 4. The driver module **9**, inside a module case **30** forming an outer shell, accom-

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modates a driver holder **40**, a driver **3**, a microphone holder **60**, and a microphone substrate **70** on which a microphone **4** is mounted, in this order from the front to the rear.

As shown in FIG. 2, the module case **30** is formed in a bottomed tubular shape that is open rearward, and a module opening **34** that communicates the inside and outside of the module case **30** is provided in the center of the bottom wall **32** that configures a front end part. The module opening **34** has substantially the same external dimensions as the housing opening **18** and is coaxially arranged. The module case **30** is formed, for example, by drawing a metal material such as iron or aluminum with a press machine. Further, an opening on the rear end side (the other end side in the axial direction) of the module case **30** is closed off by a closing member **36** made of metal. The module case **30** corresponds to a "holder insertion portion" in the present disclosure, and a driver holder **40**, which is described below, is inserted therein.

The driver holder **40** is formed in an elongated shape with the front-rear direction being its longitudinal direction. The driver holder **40** is configured by a tubular portion **42** formed in a tubular shape as a whole. In the present embodiment, the driver **3** is housed inside the tubular portion **42**. This driver holder **40** is made of an elastic material that can be elastically deformed. In the present embodiment, the driver holder **40** is made of an elastomer material such as thermoplastic elastomer (TPE) or thermoplastic polyurethane (TPU). The driver holder **40** is press-fitted from the rear side in the axial direction of the module case **30**, and is configured such that a surface portion thereof is in close contact with the inner surface of the module case **30**.

The tubular portion **42** is formed in a tubular shape whose axial direction is the front-rear direction, and is open in the front-rear direction. The tubular portion **42** is press-fitted into the module case **30** and is in close contact with the inner surface of the module case **30**, thereby forming a first sound path SP1 and a second sound path SP2.

As shown in FIGS. 3 and 4, the first sound path SP1 is configured by an opening on the front end side (the one side) of the tubular portion **42** in the axial direction. More specifically, the first sound path SP1 is configured by a through hole **44** that extends through a front wall **42A**, which covers the front end of the tubular portion **42**, in the front-rear direction. This first sound path SP1, in a state in which the tubular portion **42** is press-fitted into the module case **30**, forms a sound path that connects the driver **3** and the user's ear canal. Further, in the first sound path SP1, by adjusting the thickness dimension of the front wall **42A** and the inner diameter dimension of the through hole **44**, the volume of the first sound path SP1, and thus the volume in front of the driver, **3** can be adjusted.

The second sound path SP2 is configured by a groove portion **46** formed at a surface of the tubular portion **42**. The second sound path SP2 is communicated with the first sound path SP1 in a state in which the tubular portion **42** is press-fitted into the module case **30**, and in addition, forms a sound path extending to the microphone **4**, described below, in the module case **30**. The groove portion **46** includes a side groove portion **46A** formed in the front wall **42A** of the tubular portion **42**, a first longitudinal groove portion **46B** formed in an outer peripheral portion **42B**, and a second longitudinal groove portion **46C** formed in an extension portion **42C**, described below, of the tubular portion **42**. The side groove portion **46A** extends along the radial direction of the tubular portion **42** from a peripheral edge of the through hole **44** formed in the front wall **42A**, and is communicated with the first sound path SP1. The first

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longitudinal groove portion 46B is provided continuously with the side groove portion 46A at a radially outer end portion of the tubular portion 42, and extends along the axial direction of the tubular portion 42. The second longitudinal groove portion 46C of the extension portion 42C is provided continuously from a rear end portion of the first longitudinal groove portion 46B.

The extension portion 42C is provided at an end portion on a rear end side (the other end side) in the axial direction of the outer peripheral portion 42B. The extension portion 42C is a plate-like member that extends axially rearward from an opening portion 48 provided at the rear end portion of the outer peripheral portion 42B, and has the radial direction of the tubular portion 42 as its thickness direction. The extension portion 42C has a curved surface that is continuous with the outer peripheral portion 42B on the radial direction outer side, and closely contacts the inner surface of the module case 30. The second longitudinal groove portion 46C is formed continuously with the first longitudinal groove portion 46B at the radial direction outer side of the extension portion 42C. The second longitudinal groove portion 46C is communicated with a lateral through hole 47 passing through the extension portion 42C in the thickness direction. This second longitudinal groove portion 46C, via the lateral through hole 47, is communicated with a sound hole 4A of the microphone 4 via through holes 64A and 70A formed in a microphone holder 60 and a microphone substrate 70, which are described below.

Ribs 50 (50A, 50B, 50C) projecting toward the side of the module case 30 are integrally provided at the surface of the tubular portion 42. The ribs 50 are, for example, configured by triangular ribs, and in a state in which the tubular portion 42 is press-fitted into the module case 30, the tips of the triangular ribs are elastically deformed and closely contact the inner surface of the module case 30. That is, the ribs 50 have the function of reducing the contact area between the surface of the tubular portion 42 and the inner surface of the module case 30 and reducing frictional resistance during press fitting. Further, at the same time, they have a function of ensuring the sealing properties of the first sound path SP1 and the second sound path SP2.

Plural ribs 50A are arranged on the outer peripheral portion 42B of the tubular portion 42. The plural ribs 50A respectively extend along the axial direction of the tubular portion 42 and are arranged at a predetermined interval in the circumferential direction of the outer peripheral portion 42B. Further, a pair of ribs 50A is arranged on both sides of the first sound path SP1. While the plural ribs 50A are portions that are elastically deformed at the initial stage of press-fitting into the module case 30, since they extend along the insertion direction of the tubular portion 42, frictional resistance between the module case 30 and the tubular portion 42 is effectively reduced. Further, the ribs 50A closely contact the inner surface of the module case 30 on both sides of the first longitudinal groove portion 46B and seal the second sound path SP2. The rib 50B extends along the circumferential direction of the outer peripheral portion 42B and connects the upper ends of the ribs 50A. The rib 50B causes the outer peripheral portion 42B to closely contact the inner surface of the module case 30 along the circumferential direction. In addition, the rib 50C is arranged at the radial direction outer side of the extension portion 42C, extends so as to surround the periphery of the second longitudinal groove portion 46C, and closely contacts the inner surface of the module case 30 at the peripheral edge portion of the second longitudinal groove portion 46C.

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Further, a driver pressing portion 52 is integrally provided on the rear end side (the other end side) of the tubular portion 42 in the axial direction. The driver pressing portion 52 forms a peripheral edge portion of the opening 48 and protrudes radially inward from the outer peripheral portion 42B in an eave-like shape. In the present embodiment, a driver accommodation portion DC is provided between the front wall 42A of the tubular portion 42 and the driver pressing portion 52. The driver 3 is accommodated in the driver accommodation portion DC by surmounting the driver pressing portion 52 while elastically deforming the driver pressing portion 52 forward. In a state in which the driver 3 is thus accommodated, the inner surface of the driver accommodation portion DC closely contacts the driver 3. The inner diameter (the diameter of the inner peripheral surface) of the driver accommodation portion DC is formed to be slightly smaller than the outer diameter of the outer peripheral surface of the driver 3. Accordingly, when the driver 3 is accommodated inside the driver accommodation portion DC, the outer peripheral surface of the driver 3 and the inner peripheral surface of the driver accommodation portion DC are brought into close contact and hermetically sealed. Further, the driver pressing portion 52 presses the driver 3 forward from the rear side in the axial direction by elastic restoring force, and the front surface of the driver 3 is brought into close contact with the periphery of the first sound path SP1.

The driver 3 includes a magnetic circuit for generating an output signal, a diaphragm, and the like in a cylindrical case whose axial direction is the front-rear direction, and a well-known structure can be used as appropriate. A diaphragm of the driver 3 is arranged at the front end side of the case, and outputs an acoustic signal into the user's ear canal via the first sound path SP1.

A microphone holder 60, serving as a holder pressing portion, is arranged behind the driver holder 40. The microphone holder 60 is configured as a plate-shaped member that is bent into a substantial L-shape as a whole, and is arranged between the tubular portion 42 of the driver holder 40 and the closing member 36 in a state in which the module case 30 is closed. The microphone holder 60 is made of a resin material, for example.

The microphone holder 60 has a lower wall portion 62 that abuts on the rear surface of the driver holder 40 and a lengthwise wall portion 64 that stands axially rearward from the upper surface of the lower wall portion 62. The lower wall portion 62 abutting on the rear surface of the driver holder 40 is provided with an opening 63 that passes through the lower wall portion 62 in the axial direction (plate thickness direction) and communicates the inside of the driver accommodation portion DC with the outside thereof. This opening 63 is provided in order to release rearward pressure, generated when the diaphragm of the driver 3 vibrates, to the space rearward of the driver 3. The opening 63 also serves as a wiring opening through which wiring 72 connected to the driver 3 is inserted, as is described below. Further, the lengthwise wall portion 64 abuts on a radially inner side of the extension portion 42C of the driver holder 40. In a state in which the module case 30 is closed, the rear end of the lengthwise wall portion 64 of the microphone holder 60 is pressed by the closing member 36. As a result, the lower wall portion 62 and the lengthwise wall portion 64 are configured to press the tubular portion 42 toward the front end side in the axial direction.

A microphone fixing portion 66 is provided at the lengthwise wall portion 64 of the microphone holder 60 at an opposite surface from the surface abutting on the extension

portion 42C. A microphone substrate 70 is fixed to the microphone fixing portion 66 by a method such as fixing with a pressure-sensitive adhesive or clasp fitting. The microphone substrate 70 is arranged in a lengthwise orientation with the direction orthogonal to the axial direction of the module case 30 as the plate thickness direction, and a microphone 4 is mounted on a mounting surface facing an inner surface of the module case 30. Wiring 72 extending from the rear of the driver 3 is connected to the microphone substrate 70. The wiring 72 is connected to the microphone substrate 70 through the opening 63 formed in the lower wall portion 62 of the microphone holder 60. Here, the microphone substrate 70 is not essential in the present disclosure, and the configuration may be such that the microphone is directly fixed to the microphone fixing portion 66.

The microphone 4 is suitable for the functions of the headset 1, such as noise canceling, personal authentication, and pulse wave detection. For example, the microphone 4 is used as a microphone that collects sound in the ear canal, such as a feedback microphone for noise cancellation, or as a vibration sensor that measures vibrations in audible and non-audible frequency bands, pressure changes, or the like in the ear canal.

By mounting the microphone 4 on the lengthwise microphone substrate 70, a sound hole 4A for sound collection provided in correspondence with the inner diaphragm is opened in a direction orthogonal to the axial direction of the module case 30. The sound hole 4A, by means of through holes 64A and 70A formed in the lengthwise wall portion 64 of the microphone holder 60 and the microphone substrate 70, is communicated with the lateral through hole 47 provided in the extension portion 42C of the driver holder 40. As a result, sound (vibration) in the ear canal reaches the microphone 4 via the second sound path SP2.

The microphone 4 is arranged inside the module case 30 both in front of and behind the tubular portion 42 of the driver holder 40 along the axial direction. By arranging the driver holder 40 and the microphone 4 lengthwise in this way, the module case 30 can be made radially compact, and the entire module case 30 can be placed in the ear canal.

In the present embodiment, the microphone substrate 70 and the printed circuit board 2 described above are connected by a flexible board 6. The flexible board 6 is formed in an elongated sheet shape, and is connected to the rear end of the microphone substrate 70 as shown in FIG. 3. The other end of the flexible substrate 6 is inserted through the insertion portion 37 of the closing member 36 and protrudes outside the module case 30. The flexible substrate 6 is accommodated in a rear accommodation space of the housing 10 in a bent state (see FIG. 1).

The driver module 9 having the above-described configuration is assembled as follows. After housing the driver 3 in the driver holder 40, the microphone holder 60, the microphone substrate 70, the microphone 4, and the flexible substrate 6, which are pre-assembled in the driver holder 40, are further affixed. Then, the driver 3 and the microphone substrate 70 are connected by the wiring 72. After this, these are integrally press-fitted inside the module case 30, and by closing the rear end opening of the module case 30 with the closing member 36, assembly of the driver module 9 is completed.

In the present embodiment, a sheet-like pressure-sensitive adhesive 74 is provided between the module case 30 and the driver holder 40 and between the driver holder 40 and the microphone holder 60 to join these together. The pressure-sensitive adhesive 74 is provided in order to stabilize the positioning of the respective parts at the time of module

assembly; however, from the viewpoint of ensuring airtightness inside the module case 30, it is not essential. That is, in the present embodiment, elastic force of the driver holder 40 can be used to ensure airtightness within the module case 30. Therefore, from the viewpoint of reducing the number of parts, a configuration in which the pressure-sensitive adhesive 74 is not provided in the module case 30 may be adopted.

Mechanism and Effects

The headset 1 according to the present embodiment has been explained above; however, in a case in which the headset 1 has a noise canceling function, noise or the like that enters into the ear canal from the outside reaches the microphone 4 through the first sound path SP1 and the second sound path SP2 formed in the module case 30. Noise reaching the microphone 4 is converted into an electrical signal by the microphone 4. An electrical signal corresponding to the converted noise is input to the control unit of the printed circuit board 2, and a noise canceling signal of opposite phase is generated. Noise canceling is enabled by converting this noise cancellation signal into an acoustic signal and outputting it from the driver 3.

Moreover, in a case in which the headset 1 has a biometric verification function, using, for example, individual differences in the shape of the ear canal of an individual, an acoustic signal is output from the driver 3 and a response signal generated in the ear canal is acquired by the microphone 4. By performing frequency analysis of the response signal acquired by the microphone 4 in the control unit, the individual can be verified.

In addition, in the present embodiment, the driver holder 40 has a tubular portion 42 formed of an elastic material in a tubular shape, and the driver 3 is accommodated inside the tubular portion 42. Further, the tubular portion 42 is configured to be press-fitted into a holder insertion portion; that is, into the module case 30 inside the housing 10. As a result, the driver 3 arranged inside the housing 10 is held by the driver holder 40.

Here, the through-hole 44 provided at the front end side (one end side) of the tubular portion 42 in the axial direction configures a first sound path SP1 that communicates the driver 3 with the user's external auditory canal. As a result, an acoustic signal output by the driver 3 is radiated to the user's ear canal (outside the housing 10) via the first sound path SP1.

A second sound path SP2 is formed at the surface of the tubular portion 42. This second sound path SP2 is configured by a groove portion 46 formed at a surface of the tubular portion 42, is communicated with the first sound path SP1, and extends to the microphone 4 arranged at an outer side (rear side) of the tubular portion 42. As a result, the acoustic signal in the user's ear canal propagates to the microphone 4 via the first sound path SP1 and the second sound path SP2.

According to the configuration described above, modification of the respective sound path portions is facilitated since it suffices to change the shape of the driver holder 40 and design modification of the housing 10 is unnecessary. Further, these sound path portions (SP1, SP2) are rendered into a sealed state by press-fitting the tubular portion 42 into the module case 30. Therefore, assembly while ensuring the hermeticity of the second sound path SP2 for the microphone is facilitated. Furthermore, since the second sound path SP2 is provided integrally with the driver holder 40, the positional relationship between the driver 3 and the microphone 4 is stabilized. Therefore, variations due to assembly can be suppressed.

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Further, in the present embodiment, since a configuration is adopted in which the driver holder 40 is formed of an elastic material, the tubular portion 42 can be easily press-fitted into the module case 30 by elastic deformation thereof, and ease of assembly can be enhanced. Further, since the driver 3 is accommodated in the driver holder 40, impact absorption performance is enhanced.

In addition, in the present embodiment, ribs 50A, 50B, and 50C are formed on the surface of the tubular portion 42 and protrude from the surface, and the tubular portion 42 is press-fitted into the module case 30 while elastically deforming the ribs 50A, 50B, 50C. This reduces the frictional resistance when the tubular portion 42 is press-fitted into the module case 30, thereby improving the workability during manufacturing.

Further, in the present embodiment, the driver 3 is pressed toward the axial direction front side (the one side) of the tubular portion 42 by the elastic force of the driver pressing portion 52. As a result, the airtightness between the driver 3 and the tubular portion 42 is effectively enhanced in the vicinity of the first sound path SP1 which, in turn, contributes to an improvement in the sealing properties of the first sound path SP1. Further, in the manufacturing process, by surmounting the driver pressing portion 52, since the operator can clearly confirm that the driver 3 is housed in an appropriate position in the tubular portion 42 by visual and tactile sense, product assembly accuracy and workability are improved.

Further, in the present embodiment, since the module case 30 having a bottomed tubular shape configures the holder insertion portion, by press-fitting the driver holder 40 into the module case 30, the driver 3, the first sound path SP1 and the second sound path SP2 can be easily modularized.

Further, in the present embodiment, a microphone holder 60, as a holder pressing portion, is arranged between the closing member 36 and the tubular portion 42 of the module case 30, and presses the driver holder 40 toward the front end side (the one end side) in the axial direction. Therefore, by closing the module case 30, the tubular portion 42 can be pressed toward the front end in the axial direction, and therefore, it is possible to stabilize the positional relationship between the driver 3 and the microphone 4 and reduce variations due to assembly.

Further, in the present embodiment, since the microphone holder 60, as the holder pressing portion, is provided with the microphone fixing portion 66, a microphone 4 is arranged inside the module case 30. As a result, the main functional units of the headset 1—namely, the driver 3, the microphone 4, and the sound path—can be modularized and managed as an assembly, and therefore, product controllability is excellent. Further, by assembling the main components in advance, the installation work on the housing 10 side can be simplified. For example, in the present embodiment, the module case 30 (driver module 9) can be installed in the housing 10 simply by bonding the bottom wall 32 of the module case 30 to the bottom surface 16A of the ear canal insertion portion 16 using the pressure sensitive adhesive 74. Furthermore, by modularizing the driver 3 and the microphone 4 with the module case 30, plural products with different shapes and the like can be used together, and versatility is excellent.

Furthermore, in the present embodiment, by arranging the microphone 4 and the microphone substrate 70 with a longitudinal orientation, the microphone 4 can be arranged along the axial direction relative to the tubular portion 42. As a result, the outer dimensions of the module case 30 can be designed in accordance with the outer shape of the driver 3

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(driver holder 40), and the size of the driver module 9 is reduced in the radial direction.

In other words, in a case in which a microphone is housed with a transverse orientation inside a module case, the microphone substrate on which the microphone is mounted also has a transverse orientation (an orientation in which the axial direction is the plate thickness direction). For this reason, compared to the configuration in which the microphone is disposed longitudinally, the module case is enlarged in the radial direction in order to secure the width of the microphone substrate. In this regard, in the present embodiment, by arranging the microphone with a longitudinal orientation, it is possible to effectively suppress the module case 30 from increasing in size in the radial direction.

Further, from the viewpoint of miniaturization of the module case 30, by making the module case 30 from metal, compared to a case in which it is formed from other materials such as resin, the thickness of the side wall and bottom wall of the case can be reduced, contributing to miniaturization.

Further, in the present embodiment, since the driver module 9 is arranged inside the ear canal insertion portion 16 of the housing 10, the accommodation space for other components inside the housing 10 can be expanded. As a result, by expanding the storage space for the battery, for example, the size of the battery can be increased, and it becomes possible to increase the battery capacity of the headset 1.

Furthermore, according to the present embodiment, since the microphone 4 is arranged behind the driver 3, the microphone 4 does not interfere with the acoustic output of the driver 3, and this eliminates the drawback that the desired acoustic characteristics cannot be obtained.

Further, since the driver module 9 is arranged inside the ear canal insertion portion 16, the volume in front of the driver 3 can be reduced. Here, the volume in front of the driver is the volume of the front chamber of the driver formed between the driver and the ear canal inside the housing. That is, in the present embodiment, since the volume in front of the driver can be reduced, attenuation of high-frequency characteristics in the ear canal can be suppressed.

Second Exemplary Embodiment

The driver holder 40 according to the first embodiment described above has the second sound path SP2 formed in the overall shape of a single groove on the surface of the tubular portion 42. In contrast, as shown in FIGS. 5A-5B, the driver holder 80 according to the second embodiment is characterized in that plural groove portions 82 formed on the surface of the tubular portion 42 and communicating with each other configure the second sound path SP2. The remaining configuration is the same as in the first embodiment. Here, in the present embodiment, components that are the same as those in the first exemplary embodiment described above are appended with the same reference numerals, and explanation thereof is omitted.

The plural groove portions 82 configuring the second sound path SP2 have five side groove portions 82A formed in the front wall 42A of the tubular portion 42, and five first longitudinal groove portions 82B formed in the outer peripheral portion 42B of the tubular portion 42 and provided continuously with the side groove portions 82A. These side groove portions 82A and first longitudinal groove portions 82B have configurations corresponding to the side groove

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portion 46A and first longitudinal groove portion 46B in the first embodiment described above. Each of the five side groove portions 82A extends radially outward from the periphery of the through hole 44 formed in the center of the front wall 42A, and as viewed from the axial direction, they are arranged radially as a whole. In addition, the five first longitudinal groove portions 82B are formed continuously with the radial outer end part of each side groove portion 82A, and extend along the axial direction on the surface of the outer peripheral portion 42B. In other words, the five first longitudinal groove portions 82B are arranged at predetermined intervals along the circumferential direction of the outer peripheral portion 42B.

A circumferential groove portion 82C extending along the circumferential direction on the surface of the outer peripheral portion 42B is formed behind the five first longitudinal groove portions 82B. The circumferential groove portion 82C connects the rear end portions of the five first longitudinal groove portions 82B in the circumferential direction and is communicated with each of the first longitudinal groove portions 82B. Furthermore, one second longitudinal groove portion 82D is formed behind the circumferential groove portion 82C. The second longitudinal groove portion 82D is formed on the surface of the extension portion 42C of the tubular portion 42. The second longitudinal groove portion 82D extends axially rearward from the circumferential groove portion 82C and is communicated with the lateral through hole 47 of the extension portion 42C. The second longitudinal groove portion 82D has a structure corresponding to the second longitudinal groove portion 46C in the first embodiment described above.

In the configuration of the second sound path SP2, sound in the user's ear canal passes through the first sound path SP1, the side groove portion 82A, the first longitudinal groove portion 82B, the circumferential groove portion 82C, and the second longitudinal groove portion 82D in this order and reaches the sound hole 4A of the microphone 4.

Since the driver holder 80 having the above-described configuration essentially conforms to the configuration of the driver holder 40 according to the first embodiment, it is possible to obtain the same mechanism and effects. Further, in the present embodiment, since plural groove portions 82 (82A, 82B, 82C, 82D) are formed on the surface of the tubular portion 42 to form the second sound path SP2, these groove portions 82 enhance the impact absorption performance of the tubular portion 42. As a result, the impact resistance of the driver 3 is improved.

Further, according to the present embodiment, the cross-sectional area of the second sound path SP2 can be easily modified, and the acoustic signal of the microphone can be adjusted to desired high frequency characteristics. That is, in the second embodiment, the front wall 42A and the outer peripheral portion 42B of the tubular portion 42 are formed with five lateral groove portions 82A and five first longitudinal groove portions 82B, respectively. However, the present disclosure is not limited to this, and the cross-sectional area of the second sound path SP2 can be modified by changing the number and the cross-sectional area of each of the lateral groove portions 82A and the first longitudinal groove portions 82B as required. It is sufficient that the number of lateral groove portions 82A and first longitudinal groove portions 82B is one or more, and the number can be increased or decreased as appropriate. Further, if the cross-sectional area of the second sound path SP2 is changed, since the characteristics of the acoustic signal acquired at the position of the microphone 4 via the second sound path SP2

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change, the acoustic signal of the microphone 4 can be adjusted to desired high frequency characteristics.

Here, with reference to FIG. 6, the manner in which the frequency characteristics of the acoustic signal output from the driver 3 at the position of the microphone 4 change in accordance with a change in the second sound path SP2 is described. In each example, in a case in which an acoustic signal was output from the driver 3 in a state in which the user was wearing a headset corresponding to the respective example, the sound pressure level acquired at the position of the microphone 4 was confirmed by simulation. FIG. 6 shows the relative frequency characteristics (relative difference) of the driver sound sources at the position of the microphone 4 in Examples 2 and 3, with Example 1 as a reference. In the graph of FIG. 6, the horizontal axis indicates the frequency [Hz] of the acoustic signal output from the driver 3, and the vertical axis indicates the sound pressure level [dB] at the position of the microphone 4.

Example 1

In the headset of Example 1, one lateral groove portion 46A, one first longitudinal groove portion 46B, and one second longitudinal groove portion 46C were formed on the surface of the tubular portion of the driver holder. That is, in Example 1, the second sound path SP2 is formed in the same configuration as in the driver holder 40 of the first embodiment (see FIGS. 4A-4B).

Example 2

In the headset of Example 2, on the surface of the tubular portion of the driver holder, five lateral groove portions 82A, five first longitudinal groove portions 82B, one circumferential groove portion 82C and one second longitudinal groove portion 82D are formed. That is, in Example 2, the second sound path SP2 is formed in the same configuration as in the driver holder 80 of the second embodiment (see FIGS. 5A-5B).

Example 3

In the headset of Example 3, on the surface of the tubular portion of the driver holder, ten lateral groove portions 82A, ten first longitudinal groove portions 82B, one circumferential groove portion 82C and one second longitudinal groove portion 82D are formed. That is, in Example 3, while the basic structure is the same as in the driver holder 80 of the second embodiment, ten lateral groove portions 82A and ten first longitudinal groove portions 82B formed in the front wall 42A and the outer peripheral portion 42B of the tubular portion 42 are arranged at equal intervals along the circumferential direction to form the second sound path SP2.

As shown in FIG. 6, by changing the shape of the second sound path portion formed in the driver holder, the frequency characteristics at the position of the microphone 4 can be adjusted in the high frequency range. As a result, according to the second embodiment described above, the acoustic signal of the microphone can be adjusted to desired high-frequency characteristics.

Further, by providing plural groove portions 82 on the surface of the tubular portion 42 as in the second embodiment described above, the cross-sectional area of the second sound path SP2 can be easily secured. Accordingly, as such, by reducing the height dimension (the dimension in the radial direction) of a single groove portion 82, the outer

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diameter dimension of the tubular portion **42** can be set even smaller, and the size of the module case **30** can be reduced in the radial direction.

Third Exemplary Embodiment

In a third embodiment, as shown in FIGS. **7** and **8**, a microphone fixing portion **96** is provided integrally with an extension portion **94** of a driver holder **92**. That is, a characteristic feature is that a configuration corresponding to the driver holder **40** and the microphone fixing portion **66** of the microphone holder **60** in the first embodiment is integrally formed. Further, a characteristic feature is that a closing member **98** is provided with a holder pressing portion **98B**. The remaining configuration is the same as in the first embodiment. Here, in the present embodiment, components that are the same as those in the first exemplary embodiment described above are appended with the same reference numerals, and explanation thereof is omitted.

In the present embodiment, inside the module case **30** forming the outer shell of the driver module **90**, a driver holder **92**, a driver **3**, and a microphone substrate **70** on which a microphone **4** is mounted, are accommodated in this order from the front to the rear. Further, a rear end opening of the module case **30** is closed with a closing member **98** made of resin.

The driver holder **92** is integrally provided with an extension portion **94** on the axial direction rear end side (the other end side) of the tubular portion **42**. The extension portion **94** is a plate-like member that forms part of the tubular portion **42** and extends rearward from the axial rear end portion of the outer peripheral portion **42B**. As with the extension portion **42C** of the first embodiment, the extension portion **94** has a second longitudinal groove portion **46C** forming the second sound path **SP2** at the radially outer surface. However, a microphone fixing portion **96** is provided on the radially inner surface of the extension portion **94**, and a microphone substrate **70** is secured by a method such as pressure-sensitive adhesive or clasp-fitting. Further, the extension portion **94** is provided with a side portion through hole **95** that causes the second longitudinal groove portion **46C** and the sound hole **4A** of the microphone **4** to be communicated with each other.

The closing member **98** has a lid portion **98A** that closes the rear end opening of the module case **30** and a holder pressing portion **98B** extending axially forward from a case inner surface of the lid portion **98A**. The holder pressing portion **98B** is formed in a block shape, and the front end part thereof presses the tubular portion **42** axially forward in a state in which the module case **30** is closed. Further, the radially inside side surface of the holder pressing portion **98B** presses the microphone **4** and the microphone substrate **70** against the extension portion **94** via a cushion material **100** and supports the extension portion **94**. As a result, the positions of the driver **3** and the microphone **4** are stabilized by the closing member **98**, and efficiency of assembly is improved. In addition, it is possible to improve the airtightness of the first sound path **SP1** and the second sound path **SP2**.

Since the driver holder **92** having the above-described configuration also essentially conforms to the configuration of the driver holder **40** according to the first embodiment, it is possible to obtain the same mechanism and effects. Further, in the present embodiment, by providing the microphone fixing portion **96** integrally with the driver holder **92**, since the configuration is such that a microphone holder configured separately from the driver holder is not required,

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the number of parts and assembly man-hours for the driver module **90** can be reduced. Further, since the closing member **98** is integrally provided with the holder pressing portion **98B**, the positions of the driver **3** and the microphone **4** are stabilized. As a result, the ease of assembly of the driver module **90** can be improved.

Fourth Exemplary Embodiment

A driver module **110** of the fourth embodiment shown in FIGS. **9** to **11** is characterized in that the microphone **4** is arranged at the side of the driver **3** inside a module case **112**. Here, in the fourth embodiment, components that are the same as those in the first exemplary embodiment described above are appended with the same reference numerals, and explanation thereof is omitted.

The module case **112** is formed flat in the left-right direction (direction orthogonal to the axial direction), and is formed in a bottomed tubular shape that is open rearward in the axial direction. A module opening **34** that communicates the inside and outside of the module case **112** is provided at one end side in the left-right direction of the bottom wall **114** of the module case **112**. Further, a tubular driver holder **116** formed flat in the left-right direction is press-fitted inside the module case **112**.

The driver holder **116** is formed in a tubular shape with one end side in the left-right direction being open to the front and rear in the axial direction, and the inner side thereof configures a driver accommodation portion **DC**. The driver **3** is inserted into the driver accommodation portion **DC** through an opening **48** at the rear end side in the axial direction, which is provided so as to correspond to the driver accommodation portion **DC**. Further, a driver pressing portion **52** is provided integrally with the opening **48**. Further, an opening at the front end side in the axial direction provided so as to correspond to the driver accommodation portion **DC** is configured by a through hole **44** that penetrates the front wall **116A** of the driver holder **116** in the axial direction, and the through hole **44** configures the first sound path **SP1**.

Further, at an outer peripheral portion **116B** of the driver holder **116** at the other end in the left-right direction, a recessed accommodation portion serving as a microphone fixing portion **118** is formed. A microphone substrate **70** on which the microphone **4** is mounted is fixed to the microphone fixing portion **118**. Further, the microphone substrate **70** and the driver **3** described above are connected by wiring **72** extending in the left-right direction behind the driver holder **116**. One end of the flexible substrate **6** is connected to the microphone substrate **70**. The other end of the flexible substrate **6** is inserted through the insertion portion **37** of a closing member **120** that closes the module case **112** from behind, and protrudes to the exterior of the module case **112**. Here, the configuration may be such that the microphone substrate **70** is not provided, and the microphone is directly fixed to the microphone fixing portion **118**.

As shown in FIGS. **11A** and **11B**, at the bottom surface of the microphone **4**, a sound hole **4A** that opens toward the side of the microphone fixing portion **118** is formed. The sound hole **4A** is communicated with the second sound path **SP2** via a through hole **70A** penetrating the microphone substrate **70** in the plate thickness direction.

The second sound path **SP2** is formed integrally with the driver holder **116**. The second sound path **SP2** includes one lateral groove portion **122A** formed at the surface of the front wall **116A** of the driver holder. The lateral groove portion **122A** extends from a peripheral portion of the

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through hole **44** toward the other end side in the left-right direction. Owing to the driver holder **116** being press-fitted into the module case **112** and the front wall **116A** of the driver holder **116** abutting on the bottom wall **114** of the module case **112**, the lateral groove portion **122A** forms a sound path that is communicated with the first sound path SP1. Further, the second sound path SP2 includes a longitudinal hole portion **122B** provided continuously with the other end portion in the left-right direction of the lateral groove portion **122A**, and a vertical hole portion **122C** provided continuously with the rear end portion of the longitudinal hole portion **122B**. The longitudinal hole portion **122B** extends axially rearward from the front wall **116A** of the driver holder **116** and is connected to a lower end of the vertical hole portion **122C**. The vertical hole portion **122C** penetrates the driver holder **116** toward the side of the microphone fixing portion **118** and is communicated with the sound hole **4A** of the microphone **4**.

According to the foregoing configuration, the second sound path SP2 is configured by the lateral groove portion **122A**, the longitudinal hole portion **122B**, and the vertical hole portion **122C**, and extends from the first sound path SP1 to the microphone **4**. As a result, sound outside the module case **112** reaches the microphone **4** via the first sound path SP1 and the second sound path SP2.

In the present embodiment, as in the first embodiment, since the second sound path SP2 for the microphone is formed by press-fitting the driver holder **116** inside the module case **112**, it is possible to easily modify the second sound path SP2 and ensure airtightness, and to improve the ease of assembly.

Further, in the present embodiment, since the microphone **4** is arranged at the side of the driver **3** inside the module case **112**, the module case **112** and the driver holder **116** can be made more compact in the axial direction.

Fifth Exemplary Embodiment

While a driver module **130** of a fifth embodiment shown in FIGS. **12A** and **12B** essentially conforms to the configuration of the driver module **110** of the fourth embodiment, the configuration differs in that the second sound path SP2 and the first sound path SP1 are not communicated with each other. Here, in the present embodiment, components that are the same as those in the fourth exemplary embodiment are appended with the same reference numerals, and explanation thereof is omitted.

The module case **112** of the driver module **130** has a module opening **34** formed at one end side of the bottom wall **114** in the left-right direction, and a sound path module opening **132** is formed at the other end side in the left-right direction. The sound path module opening **132** is configured by a through hole penetrating the bottom wall **114** in the plate thickness direction, and communicates the outside of the module case **112** with the second sound path portion P2.

The second sound path SP2 is provided to the side of the first sound path SP1 for the driver, and is configured by a longitudinal hole portion **122B** and a vertical hole portion **122C** which penetrate the front wall **116A** and the outer peripheral portion **116B** of the driver holder **116**.

Since the configuration of the fifth embodiment described above essentially conforms to the configuration of the fourth embodiment, a similar mechanism and effects can be obtained. Further, at the surface of the driver holder **116**, since the surface shape of the front wall **116A**, which affects the sealing of the first sound path SP1 and the second sound path SP2, can be simplified, the contact surface with the

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module case **112** is stabilized, and the sealing performance of the respective sound path portions is improved.

Sixth Exemplary Embodiment

A driver module **200** according to a sixth embodiment is explained below with reference to FIGS. **13** and **14**. Here, in the present embodiment, components that are the same as those in the first exemplary embodiment described above are appended with the same reference numerals, and explanation thereof is omitted. The driver module **200** according to this sixth embodiment is characterized in that two microphones **400A**, **400B** can be accommodated inside the module case **210**.

As shown in FIG. **13**, inside the module case **210**, a driver holder **220**, a driver **3**, a first microphone holder **230**, a first microphone substrate **70A** on which the first microphone **400A** is mounted, and a second microphone substrate **70B** on which a second microphone holder **240** and the second microphone **400B** are mounted, are housed in this order from the front to the rear.

The module case **210** is formed in a bottomed tubular shape that is open to the rear in the axial direction, and a module opening **214** that communicates the inside and outside of the module case **210** is provided at the center of the bottom wall **212** of the module case **210**. Further, the module case **210** has a sound path accommodation portion **216** formed by protruding a portion of the outer periphery radially outward.

The driver holder **220** is formed in a tubular shape that is open axially to the front and rear, and is press-fitted from the rear opening of the module case **210**. The driver holder **220** essentially conforms to the configuration of the driver holder **40** of the first embodiment, and has a cylindrical tubular portion **222** and an extension portion **224** integrally provided at a rear end portion of the tubular portion **222**. The first sound path SP1 is formed at a front wall **222A** of the tubular portion **222** by a through hole **44** formed therethrough. The second sound path SP2 is formed by a groove portion **46** formed at the surface of the tubular portion **222** and the extension portion **224** so as to communicate with the first sound path portion. Since the configurations of the first and second sound path portions SP1 and SP2 are the same as in the first embodiment, detailed description thereof is omitted.

The driver holder **220** is formed so that the outer peripheral portion at which the second sound path SP2 is formed protrudes in the radial direction and is thicker than other portions. This thick portion is housed in the sound path accommodation portion **216** of the module case **210** when the driver holder **220** is press-fitted into the module case **210**. The driver holder **220**, by having the second sound path SP2 formed thickly and portions other than the second sound path SP2 made thinner, has a reduced outer diameter and a reduced size. A rib **50** surrounding the second sound path SP2 is further formed at the surface of the driver holder **220**, and the leading ends of the ribs **50** are elastically deformed and come into close contact with the inner surface of the sound path accommodation portion **216**. As a result, the second sound path SP2 is formed with excellent sealing properties.

As shown in FIG. **14**, inside the driver accommodation portion DC formed inside the tubular portion **222** of the driver holder **220**, the driver **3** is inserted from the opening **226** at the rear end side of the tubular portion **222**, and the first microphone holder **230** is disposed behind the driver **3**. The first microphone holder **230** essentially conforms to the configuration of the microphone holder **60** of the first

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embodiment, and a lower wall portion **232** abutting on the rear surface of the driver holder **220**, and a longitudinal wall portion **234** standing axially rearward from the upper surface of the lower wall portion **232**, form a plate-like member bent in a substantial L-shape.

The lower wall portion **232** has an opening **236** penetrating in the axial direction (thickness direction) thereof, and wiring **72** connected to the rear surface of the driver **3** is drawn rearward through the opening **236**. The extension portion **224** of the driver holder **220** abuts on an outer surface of the longitudinal wall **234** facing the inner surface of the module case **210**, the longitudinal wall **234** forming a curved surface that is substantially flush with the extension portion **224**. A first microphone fixing portion **66A** is provided at an inner surface of the longitudinal wall portion **234** provided on the opposite side from this outer surface. A first microphone substrate **70A** is fixed to the first microphone fixing portion **66A** using a pressure-sensitive adhesive.

The first microphone substrate **70A** is arranged in a longitudinal orientation similarly to the microphone substrate **70** of the first embodiment, and the first microphone **400A** is mounted on a mounting surface of the first microphone substrate **70A** at an opposite side from the first microphone fixing portion **66A**. The sound hole of the first microphone **400A** is communicated with the second sound path **SP2** via a through hole formed penetrating each of the first microphone substrate **70A**, the longitudinal wall portion **234** of the first microphone holder **230**, and the extension portion **224** of the driver holder **220**. As a result, sound (vibration) in the ear canal reaches the first microphone **400A** via the second sound path **SP2**. The first microphone **400A** is used, for example, for feedback-type noise cancellation. The first microphone substrate **70A** is connected to an elongated sheet-like first flexible board **6A**, and is electrically connected to the printed circuit board **2** inside the housing **10** via a connector **5** provided at the leading end of the first flexible circuit board **6A**.

A second microphone holder **240** is disposed behind the first microphone holder **230**. As with the first microphone holder **230**, the basic component elements of the second microphone holder **240** also conform to the configuration of the microphone holder **60** of the first embodiment, and a lower wall portion **242** and a longitudinal wall portion **244** form a plate-like member bent in a substantial L-shape.

The second microphone holder **240** is arranged in a reversed orientation from that of the first microphone holder **230** in the front-rear direction, and is housed inside the module case **210** in an inverted posture relative to the first microphone holder **230**. In this state, the longitudinal wall portions **234** and **244** of the first and second microphone holders **230** and **240** are arranged facing each other inside the module case **210**. Further, the lower wall portion **242** of the second microphone holder **240** abuts on the rear surface of the longitudinal wall portion **234** of the first microphone holder **230**, and the opening on the rear end side of the module case **210** is closed off by the lower wall portion **242**. In this lower wall portion **242**, two insertion portions **246** are formed for insertion of the first flexible substrate **6A** connected to the first microphone substrate **70A** and of a second flexible substrate **6B** connected to the second microphone substrate **70B**, which is described below.

A sheet-like gasket **250** abuts on the outer surface of the longitudinal wall portion **244** facing the inner surface of the module case **210**, and forms a curved surface that is substantially flush with the outer surface of the longitudinal wall portion **244**. This gasket **250** is formed using a similar elastic material to the driver holder **220**, and in the present embodi-

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ment, an elastomer material such as TPE or TPU is used, as an example. This gasket **250** is disposed in a state of being press-fitted between the longitudinal wall portion **244** of the second microphone holder **240** and the module case **210**.

Further, by bringing the respective faces of the gasket **250** into close contact with the outer surface of the longitudinal wall portion **244** and the inner surface of the module case **210**, a sealed third sound path **SP3** is formed, communicating the inside and outside of the module case **210**. The third sound path **SP3** is configured by a through hole **252** formed penetrating the gasket **250**. By arranging the through hole **252** coaxially with the through hole **218** formed at the outer periphery of the module case **210**, at an outer side the user's ear canal, the inside and outside of the module case **210** are communicated with each other. Here, a rib **50** surrounding the third sound path **SP3** is formed at the surface of the gasket **250**, similarly to in the driver holder **220**, and the leading ends of the rib **50** are elastically deformed and adhere to the inner surface of the module case **210**.

A second microphone fixing portion **66B** is provided at an inner surface of the longitudinal wall portion **244** provided at an opposite side from the outer surface of the second microphone holder **240**. A second microphone substrate **70B** is fixed to the second microphone fixing portion **66B** using a pressure-sensitive adhesive. The second microphone substrate **70B** is arranged in a longitudinal orientation similarly to the first microphone substrate **70A**, and is disposed at the rear side of the first microphone substrate **70A**. The second microphone **400B** is mounted on a mounting surface of the second microphone substrate **70B** at an opposite side from the second microphone fixing portion **66B**. The sound hole of the second microphone **400B** is communicated with the third sound path **SP3** via through holes (reference numerals are omitted for each of these) formed through the longitudinal wall portion **244** of the second microphone substrate **70B** and the second microphone holder **240**, respectively. As a result, sound (vibration) outside the ear canal reaches the second microphone **400B** via the second sound path **SP3**. The second microphone **400B** is used, for example, for feed-forward-type noise cancellation. The second microphone substrate **70B** is connected to an elongated sheet-like second flexible board **6B**, and is electrically connected to the printed circuit board **2** inside the housing **10** via a connector **5** provided at the leading end of the second flexible circuit board **6B**.

In the assembly of the driver module **200** according to the sixth embodiment described above, the driver **3** is housed in the driver holder **220** in advance, and the assembly begins in a state in which the first microphone substrate **70A**, the first microphone **400A** and the first flexible board **6A** are fixed to the first microphone holder **230**. Further, the driver holder **220** and the first microphone holder **230** are fixed using the pressure sensitive adhesive **74**, and the assembly is press-fitted into the module case **210**. After this, the second microphone holder **240**, in a state in which the second microphone substrate **70B**, the second microphone **400B**, the second flexible board **6B** and the gasket **250** are fixed in advance thereto, is inserted into the module case **210**, and by closing the opening at the rear end side of the module case **210** with the second microphone holder **240**, assembly is completed. Here, the pressure sensitive adhesive **74** is also arranged between the module case **210** and the driver holder **220**.

Since the driver module **200** having the foregoing configuration essentially conforms to the configuration of the driver module **9** of the first embodiment, a similar mechanism and effects can be obtained. Further, in the present

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embodiment, the first microphone **400A** used for feedback-type noise canceling and the second microphone **400B** used for feedforward-type noise canceling are modularized integrally with the driver **3**, and the noise canceling performance can be improved.

Further, according to the present embodiment, the first microphone holder **230** and the second microphone holder **240** are configured by approximately L-shaped plate-shaped members arranged in reversed orientations, and are accommodated in the module case **210** in mutually inverted postures. As a result, the accommodation space for the first and second microphones **400A**, **400B** can be reduced in the axial direction, and the size of the module case **210** can be reduced. Here, in the embodiment described above, the first microphone **400A** and the second microphone **400B** are arranged next to each other in the front-rear direction in the module case **210**; however, there is no limitation to this, and it is also possible to make the module case **210** more compact in the axial direction by adopting a configuration in which the first microphone **400A** and the second microphone **400B** are arranged so as to face each other.

Further, in the embodiment described above, the first microphone **400A** is used for noise cancellation; however, there is no limitation to this, and it may be used as a vibration sensor for measuring vibrations in audible and non-audible frequency bands, pressure changes, or the like, in the ear canal.

Seventh Exemplary Embodiment

A driver module **300** according to a seventh embodiment and a headset **302** using the driver module **300** are explained below with reference to FIGS. **15** to **19**. Here, in the present embodiment, components that are the same as those in the first exemplary embodiment described above are appended with the same reference numerals, and explanation thereof is omitted. The driver module **300** according to the seventh embodiment is characterized in that a part of the module case **310** is exposed from the housing **304**.

As shown in FIG. **15**, the headset **302** has a hollow housing **304** accommodating functional components thereinside. The housing **304** is formed by fitting together a front housing **304A** and a rear housing **304B**. The front housing **304A** is formed in a tubular shape having an oblique truncated cone outer shape as a whole, and a housing opening **306** that communicates the inside and outside of the front housing **304A** is provided at an apex of the oblique truncated cone. A driver module **300** is inserted into this housing opening **306**. The rear housing **304B** is formed in the shape of a shallow dish that opens forward, and is disposed so as to close off the rear opening of the front housing **304A**.

A box-shaped inner housing **308** opening forward is accommodated in an accommodation space formed by the front housing **304A** and the rear housing **304B**, and a battery **8** is held inside the inner housing **308**. Further, a first printed circuit board **2A** is arranged at a front side of the inner housing **308** in the accommodation space, and a second printed circuit board **2B** is arranged at a rear side of the inner housing **308**. The first and second printed circuit boards **2A** and **2B** are arranged in an orientation in which the board thickness direction is substantially the front-rear direction, and electronic components necessary for controlling the headset **302** are mounted on respective mounting surfaces. At the first printed circuit board **2A** arranged at the front side of the inner housing **308**, a charging terminal **7** (not shown

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in FIG. **15**) of the headset **302**, and a substrate assembly **330** connected to the driver module **300** via a connector **5** are connected.

At the housing opening **306** formed at the front end of the front housing **304A**, a module case **310** forming an outer shell of the driver module **300** is inserted. The module case **310** is formed in a bottomed tubular shape that is open rearward, and a module opening **314** that communicates the inside and outside of the module case **310** is provided in the center of the bottom wall **312** that configures the front end part. The leading end of the module case **310** protrudes forward from the front housing **304A** and is disposed in the user's ear canal.

Here, an earpiece **309** is attached to the leading end (front end) of the module case **310**. The earpiece **309** has a cylindrical portion **309A** fitted to the outer periphery of the module case **310**, and a hemispherical cover-like contact portion **309B** integrally provided at the leading end of the cylindrical portion **309A**. A first groove portion **316** annularly formed along the circumferential direction is formed at the leading end of the module case **310**, and the earpiece **309** is attached to the module case **310** by fitting a fitting projection **309C**, provided at the inner circumference of the cylindrical portion **309A** of the earpiece **309**, into the first groove portion **316**.

As shown in FIG. **16**, an annular second groove portion **318** is formed at an intermediate portion of the module case **310**. The second groove portion **318** is formed with a vent hole **319** (see FIG. **17**) that communicates the inside and outside of the module case **310**, and the driving of the driver **3** is optimized by adjusting the air pressure inside the module case **310**. An annular vent cover **322** made of a mesh material is attached to the second groove portion **318**, which prevents intrusion of external foreign matter through the vent hole **319**. Furthermore, an annular third groove portion **320** is formed at the rear end portion of the module case **310**. An annular third substrate portion **336** that configures a substrate assembly **330**, which is described below, is mounted in the third groove portion **320**.

As shown in FIG. **17**, inside the module case **310**, a driver holder **40**, a driver **3**, a microphone holder **60**, and the substrate assembly **330** on which a microphone **4** is mounted, are accommodated in this order from the front to the rear, and the rear opening of the module case **310** is closed off by a closing member **350**. Since the configurations of the driver holder **40**, the driver **3**, and the microphone holder **60** are similar to those of the first embodiment, detailed explanation is omitted.

As shown in FIG. **18**, the substrate assembly **330** is formed by connecting a plate-like printed board and a sheet-like flexible board, and is provided with a first substrate portion **332**, a second substrate portion **334** and a third substrate portion **336**. The first substrate portion **332** is configured by a plate-shaped printed circuit board, and is fixed to the microphone fixing portion **66** of the microphone holder **60**. The first substrate portion **332** is arranged in the module case **310** in a longitudinal orientation, similarly to the microphone board **70** of the first embodiment. The microphone **4** is mounted on a mounting surface of the first substrate portion **332** at an opposite side from the microphone fixing portion **66**. A sound hole of the microphone **4** is communicated with the second sound path **SP2** via through holes (not provided with reference numerals) formed penetrating each of the first substrate portion **332**, the longitudinal wall portion **64** of the first microphone holder **60**, and the extension portion **224** of the driver holder **40**. A thermistor **340** for measuring the ambient temperature

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of the driver module **300** is mounted on the mounting surface of the first substrate portion **332** in addition to the microphone **4**.

The second substrate portion **334** is formed with an elongated sheet shape extending from an end portion of the first substrate portion **332** along the axial direction of the module case **310**, and is configured by a flexible substrate. The leading end of the second substrate portion **334** is electrically connected to the first printed board **2A** inside the housing **304** via the connector **5**.

The third substrate portion **336** is formed with an elongated sheet shape extending from an end portion of the first substrate portion **332** along the circumferential direction of the module case **310**, and is configured by a flexible substrate. The third substrate portion **336** has a pad, which can be used for a capacitive-sensing-method proximity sensor, for example. In such a case, by forming the module case **310** from a metal material, the capacitance can be further increased, and the performance of the proximity sensor can be improved. The third substrate portion **336** is drawn out to the exterior through a slit **324** (see FIG. **19**) formed in the outer periphery of the module case **310**, and is mounted in the annular third groove portion **320**.

As shown in FIG. **19**, a closing member **350** is arranged at the rear side of the microphone holder **60**. The closing member **350** has a lid portion **350A** that closes off the opening at the rear end side of the module case **310**, and a holder pressing portion **350B** erected on the front side in the axial direction from a surface at a case inner side of the lid portion **350A**. Since the basic configuration of the closing member **350** conforms to the configuration of the closing member **98** according to the third embodiment, detailed explanation is omitted; however, in a state in which the module case **310** is closed by the lid portion **350A**, the longitudinal wall portion **64** of the microphone holder **60** and the holder pressing portion **350B** are arranged facing each other inside the case, and the first substrate portion **332** provided with the microphone **4** is arranged between the longitudinal wall portion **64** and the holder pressing portion **350B**. The holder pressing portion **350B** presses the longitudinal wall portion **64** of the microphone holder **60** via the first substrate portion **332** with a lateral direction (radial direction) force, and the driver holder **40** is brought into close contact with the module case **310**. Further, the second substrate portion **334** of the substrate assembly **330** is drawn out to the exterior of the module case **310** through an insertion portion **352** formed through the lid portion **350A**.

Assembly of the driver module **300** according to the seventh embodiment begins in a state in which the driver **3** is housed in the driver holder **40** in advance, and the first substrate portion **332** of the substrate assembly **330** is fixed to the microphone holder **60**. Further, the driver holder **40** and the microphone holder **60** are fixed using the pressure sensitive adhesive **74**, and the assembly is press-fitted into the module case **310**. At this time, the third substrate portion **336** of the substrate assembly **330** is drawn out from the slit **324** of the module case **310**, and is wound onto the third groove portion **320** at the outer periphery of the module case **310**. After this, the opening on the rear end side of the module case **310** is closed off by the closing member **350** and the assembly is complete. At the ring-shaped pressure-sensitive adhesive **74** that bonds the driver holder **40** and the microphone holder **60**, a front mesh **74A** is provided that covers the module opening **314** of the module case **310** and prevents waste matter and the like from entering. Further, the pressure sensitive adhesive **74** is also disposed between

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the module case **310** and the driver holder **40** and between the microphone holder **60** and the closing member **350**.

Since the driver module **300** having the foregoing configuration essentially conforms to the configuration of the driver module **9** of the first embodiment, a similar mechanism and effects can be obtained. Further, in the present embodiment, a portion of the driver module **300** is arranged outside of the housing **304**, and a leading end side exposed from the housing **304** is inserted into the user's ear canal. That is, the module case **310** configures the ear canal insertion portion of the headset **302**. This makes it possible to easily meet the demand for a reduced diameter of the ear canal insertion portion of the headset **302**.

Supplementary Explanation

While examples of embodiments of the present disclosure have been explained above, the present disclosure can be modified as appropriate without departing from the gist of the invention, and omission, replacement, and modification of the configurations in the respective embodiments described above can be performed.

For example, in the foregoing embodiments, the driver holders **40**, **80**, **92**, **116**, **220** and the gasket **250** are formed with an elastomer material; however, the present disclosure is not limited to this, and any elastic material that can be elastically deformed may be used. For example, they may be made of silicone rubber. Further, in a case in which the configuration of the present disclosure is a driver holder made of conductive silicone in which a conductive filler is mixed into silicone rubber, it is possible to prevent the generation of static electricity when assembling the driver module and when using the headset.

Further, although the ribs **50A**, **50B**, and **50C** are configured by triangular ribs in each of the foregoing embodiments, the present disclosure is not limited to this. The shape of the ribs may be hemispherical or trapezoidal.

Further, in the first to sixth embodiments described above, the driver modules **9**, **90**, **200** are arranged inside the ear canal insertion portion **16**; however, the present disclosure is not limited to this. The driver modules **9**, **90** may be arranged in the eardrum-side portion of the housing **10**; for example, behind the ear canal insertion section **16**. Alternatively, as in the driver module **300** of the seventh embodiment, a part of the driver module **300** may be exposed outside of the housing **304**.

Further, in each of the foregoing embodiments, the driver **3** and the microphones **4**, **400A**, **400B** are housed in the module case **30**; however, the present disclosure is not limited to this. Parts other than the driver **3** and the driver holder **40**, such as the microphone **4**, may be arranged outside the module case **30**. Further, the microphone **4** is not limited to a configuration arranged at the rear side of the driver **3**, and may be arranged at a side of the driver.

Further, in each of the above-described embodiments, the holder insertion portion is configured by the module case **30**; however, the present disclosure is not limited to this. The holder insertion portion may be configured so as to be provided in the housing **10**. For example, the ear canal insertion portion **16** may be configured as a holder insertion portion, and the driver holder **40** may be directly press-fitted into the ear canal insertion portion **16**. Alternatively, behind the ear canal insertion portion **16** inside the housing **10**, a bottomed tubular accommodation portion communicated with the ear canal insertion portion may be integrally formed, and the accommodation portion may be used as the holder insertion portion.

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Further, the housings **10, 304** in each of the embodiments described above are merely examples, and any housing that accommodates components of the headset at the interior thereof may be used. The shape and size of the housing can be changed in various ways. Further, although the headset of the foregoing embodiments is a wireless headset, the present disclosure is not limited to this, and the headset may be connected to a terminal via a wire. Further, the headset may be for both ears or for one ear. Further, the driver module and driver holder of the foregoing embodiments may be applied to wearable devices.

The disclosure of Japanese Patent Application No. 2020-170811 filed on Oct. 8, 2020, is incorporated herein by reference in its entirety.

All publications, patent applications and technical standards mentioned herein are incorporated herein by reference to the same extent as if each individual publication, patent application and technical standard were specifically and individually indicated to be incorporated by reference.

The invention claimed is:

1. A driver module comprising:
 - a driver;
 - a microphone having a sound hold for sound collection;
 - a driver holder for holding the driver at an interior of a hollow housing, the driver holder comprising:
 - a tubular portion formed in a tubular shape, the driver being accommodated at an inner side thereof, and the tubular portion being configured to be press-fitted into a holder insertion portion disposed inside the housing,
 - a first sound path configuring an opening at one end side in an axial direction of the tubular portion, and communicating the driver with an exterior of the driver holder, and
 - a second sound path configured by a groove formed at a surface of the tubular portion, the second sound path being communicated with the first sound path and extending as far as the microphone disposed at an outer side of the tubular portion,
 - wherein the microphone is disposed at a rear side with respect to the driver, such that the sound hold is positioned within a range at which the tubular portion is formed when seen in front-rear direction.
2. The driver module of claim 1, wherein the driver module is formed with an elastic material.
3. The driver module of claim 2, wherein:
 - a rib projecting from the surface is formed at the surface of the tubular portion, and the tubular portion is press-fitted into the holder insertion portion while elastically deforming the rib.
4. The driver module of claim 2, wherein:
 - a driver pressing portion is provided at the tubular portion, the driver pressing portion protruding toward an inner side in a radial direction from an opening at another end side in the axial direction at which the driver is inserted, and
 - the driver is accommodated inside the tubular portion by surmounting the driver pressing portion, and is pressed toward one side in the axial direction by elastic force of the driver pressing portion.
5. The driver module of claim 2, wherein a plurality of the grooves is formed at the surface of the tubular portion.
6. The driver module of claim 2, wherein:
 - the holder insertion portion is configured by a bottomed tubular module case, and
 - the tubular portion is press-fitted inside the module case.

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7. The driver module of claim 1, wherein:
 - a rib projecting from the surface is formed at the surface of the tubular portion, and the tubular portion is press-fitted into the holder insertion portion while elastically deforming the rib.
8. The driver module of claim 7, wherein:
 - a driver pressing portion is provided at the tubular portion, the driver pressing portion protruding toward an inner side in a radial direction from an opening at another end side in the axial direction at which the driver is inserted, and
 - the driver is accommodated inside the tubular portion by surmounting the driver pressing portion, and is pressed toward one side in the axial direction by elastic force of the driver pressing portion.
9. The driver module of claim 7, wherein a plurality of the grooves is formed at the surface of the tubular portion.
10. The driver module of claim 7, wherein:
 - the holder insertion portion is configured by a bottomed tubular module case, and
 - the tubular portion is press-fitted inside the module case.
11. The driver module of claim 1, wherein:
 - a driver pressing portion is provided at the tubular portion, the driver pressing portion protruding toward an inner side in a radial direction from an opening at another end side in the axial direction at which the driver is inserted, and
 - the driver is accommodated inside the tubular portion by surmounting the driver pressing portion, and is pressed toward one side in the axial direction by elastic force of the driver pressing portion.
12. The driver module of claim 11, wherein a plurality of the grooves is formed at the surface of the tubular portion.
13. The driver module of claim 11, wherein:
 - the holder insertion portion is configured by a bottomed tubular module case, and
 - the tubular portion is press-fitted inside the module case.
14. The driver module of claim 1, wherein a plurality of the grooves is formed at the surface of the tubular portion.
15. The driver module of claim 14, wherein:
 - the holder insertion portion is configured by a bottomed tubular module case, and
 - the tubular portion is press-fitted inside the module case.
16. A headset, comprising the driver module of claim 1.
17. A driver holder for holding a driver at an interior of a hollow housing, the driver holder comprising:
 - a tubular portion formed in a tubular shape, the driver being accommodated at an inner side thereof, and the tubular portion being configured to be press-fitted into a holder insertion portion disposed inside the housing;
 - a first sound path configuring an opening at one end side in an axial direction of the tubular portion, and communicating the driver with an exterior of the housing; and
 - a second sound path configured by a groove formed at a surface of the tubular portion, the second sound path being communicated with the first sound path and extending as far as a microphone disposed at an outer side of the tubular portion,
 - wherein a plurality of the grooves is formed at the surface of the tubular portion.
18. A headset, comprising the driver holder of claim 17.
19. A driver holder for holding a driver at an interior of a hollow housing, the driver holder comprising:
 - a tubular portion formed in a tubular shape, the driver being accommodated at an inner side thereof, and the tubular portion being configured to be press-fitted into a holder insertion portion disposed inside the housing;

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a first sound path configuring an opening at one end side in an axial direction of the tubular portion, and communicating the driver with an exterior of the housing; and

a second sound path configured by a groove formed at a surface of the tubular portion, the second sound path being communicated with the first sound path and extending as far as a microphone disposed at an outer side of the tubular portion,

wherein the holder insertion portion is configured by a bottomed tubular module case, and the tubular portion is press-fitted inside the module case.

20. A headset, comprising the driver holder of claim **19**.

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